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Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation

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ABSTRACT

Preserving Salience By Maintaining Perceptual Differences
for Image Creation and Manipulation

Amy Ashurst Gooch

Image creation is an attempt to record a scene for presentation to a viewer. However, there is a precarious mapping between how a scene would be perceived and how an image is perceived. For example, although the human visual system does not have a strong absolute sense of color or intensity, images are stored as a discrete set of color or intensity values at each pixel. The goal of this dissertation is to create perceptually salient images by more directly expressing perceived differences in scenes as displayed differences in images. This dissertation examines two methods of perceptual difference preservation for two long-standing problems: 1) a color to grayscale conversion algorithm that preserves all visible color changes through high-dimensional distance metrics; 2) hidden feature finding techniques that identify correspondences and mismatches between X-ray and visible light images of historically significant paintings through segmentation and local gradient statistics.

Dedication

To Bruce:

More than the sun,

the moon,

and all the stars.

Acknowledgements

Several years in the making, my dissertation is the product of no blood, but definitely sweat, and even a few tears, and could not have been done without the support of my family, friends, and peers. Thanking Bruce doesn't seem to be enough, there is nothing I could say to cover how much I enjoy working side by side with him, value his support, and the ability to preview what I am getting myself into. I would also like to thank "mini" Gooch, who I credit with my ability to handle all of the changes that accompany obtaining this final degree, as well as the new job, new home, new country. I thank Carrie, my best friend and sister, Mom and Dad, as well as Tice, Rebecca, Georgette Zubeck and the Gooches who have been through it all from nearly half way across the United States, but always right there through the web or by phone.

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CHAPTER 1

Introduction

In this dissertation, I will address how to create or extend the communication power of images by mapping visible changes from one domain to another. While the human visual system is acutely sensitive to many kinds of visible change, until the advent of computable images these changes were restricted to simple one-to-one mappings from a scene to an image that represents the scene. For example, suppose we wish to depict a beach or seaside scene, and somehow capture changes in tide levels within a single photograph. If we only use photographic film, our abilities to capture these changes are severely restricted. We could not map tide level changes to changes in position, intensity, shape, orientation, or depth of field in that film image.

Removing the requirement of one-to-one mappings enables new opportunities for effectively conveying information in images. In the scene observing the tide levels above, a single image could capture changes in tide levels by denoting the tide levels with multiple lines, color-coded by time of day. In this example, tide position in an image maps to line position and time maps to line color. In this dissertation, I will provide descriptions of several previous methods that have explored this vast domain of alternatives to traditional one-to-one mappings as well as describe and evaluate two new advances that preserve changes across and within domains such as color-to-grayscale conversion and analysis of visible light and X-ray images of

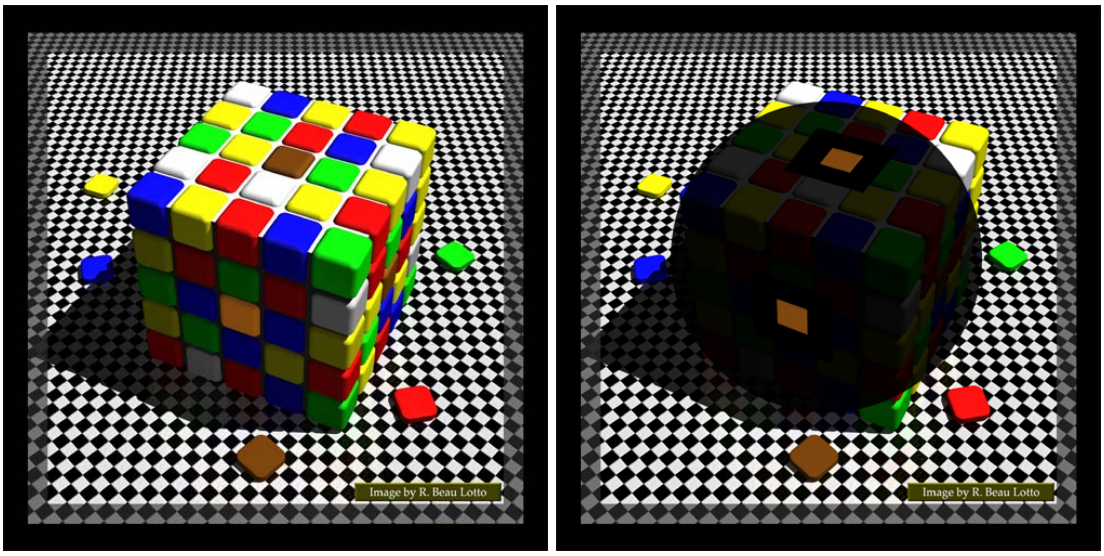


Figure 1.1. *Chrominance and luminance perception are based on relative assessments. Although the orange squares on the center top and the center front appear to have difference chrominance and luminance values (Left), their pixel values are exactly the same (Right). Images © 2006 Lotto and Purves* `http://www.lottolab.org`.

paintings. Additionally, I will discuss how user interaction is both necessary and beneficial in both these techniques.

Change sensing is an intrinsic component of vision. Accordingly, recording and conveying change should be an intrinsic part of image creation. Vision scientists hypothesize that the human visual system does not perceive absolute values, and instead chrominance and luminance perception are based upon relative assessments, in part due to the center-surround organization of cells in the early stages of visual processing [**30**], as illustrated in Figure 1.1.

I propose that preserving relationships between nearby pixels in an image is much more important than preserving the absolute pixel values themselves. Such relationships can be as simple as chrominance or luminance differences between

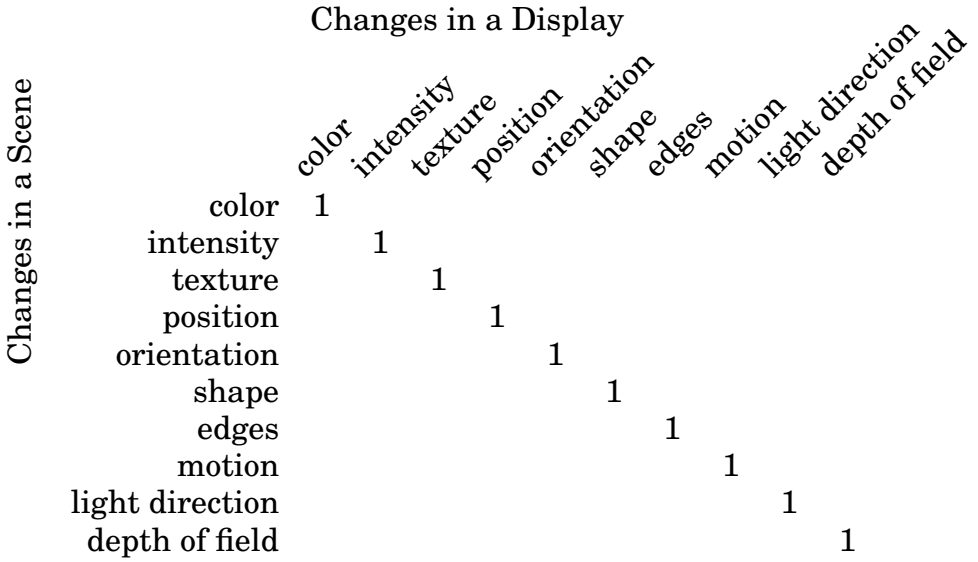


Figure 1.2. *Mapping from perceived visible parameters in a scene to a display. The "1"s represent a one-to-one mapping. I believe that as long as the magnitude of the differences in the visible parameters (i.e. color, intensity, ..., motion) are preserved, there are many mappings off the diagonal to explore.*

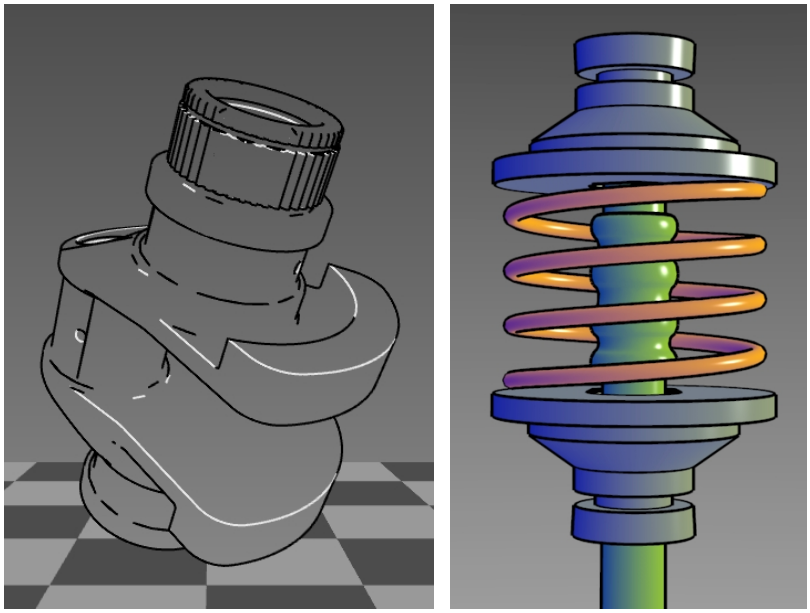
nearby pixels, or as complex and abstract as context-based information. For example, a pixel's membership in a region labeled as human, grass texture, or convex shadowed shape should affect any changes or manipulations to that pixel or its neighbors in order preserve contextual or semantic relationships. Based upon the idea that difference-preserving mappings enable the preservation of salient or important features, this dissertation will explore mappings that preserve the semantic information contained in imagery even if pixel values of color and intensity from the scene are not strictly preserved.

Typically, computer graphics algorithms map the color and intensity viewed in a 3D scene to a pixel value on a display device with no other coupling between pixel values. In order to take a more general view and consider the set of all potential

mappings, converting from changes in a scene to changes in an image for display can be organized as a matrix of mappings, as shown in Figure 1.2. The potential mappings in the matrix of Figure 1.2 include a partial set of the visible parameters sensed by the human visual system [**38**], as well a few common rendering parameters. The matrix is not complete; many more parameters could be included such as viewpoint, illumination, transparency, and reflectance.

The main diagonal of this matrix represents the ideal of photorealistic rendering: every scene change is faithfully reproduced on the display. Photorealistic rendering copies changes in a scene to changes in a display, discarding other possibilities that could be equally or potentially more effective at communicating content and context, such as the technical illustrations shown in Figure 1.3. My research is an attempt to create effective imagery by exploring mappings off the main diagonal of this matrix.

The notion of images as a mapping of a view of the world is not new. Willats [**50**] developed a structural language to describe "how" pictures such as a work of fine art, a child's drawing, a traffic sign, or a repair manual are different. In "An Invitation to Discuss Computer Depiction", Durand [**13**] generalized Willat's approach for basic depiction techniques, describing four kinds of system mappings: spatial, primitive, attributes, and marks. The *spatial* system, with respect to traditional computer graphics techniques, typically maps 3D spatial properties to 2D spatial properties. The *primitive* system maps object space primitives (points, lines, surfaces, volumes) to picture space primitives (points, lines, regions). The *attribute* system maps visual properties such as color, texture, and orientation to picture



(a) Edges to lines (b) Lighting to Color

Figure 1.3. Examples of visually effective change-based mappings across different domains.

primitives. The *mark* system places *primitives* at their *spatial* location with the corresponding *attributes*. The goals of Durand and the work presented in this dissertation for optimizing image creation for the sake of communication are the same. However, I claim that although it is important to understanding and use these mappings carefully, enforcing mappings that preserve the *changes* they represent is of primary importance for maximizing communication content of images.

The computer graphics literature and the images around us provide many examples of effective imagery or imaging techniques that fit within the paradigm of exploring off-diagonal change-based mappings.

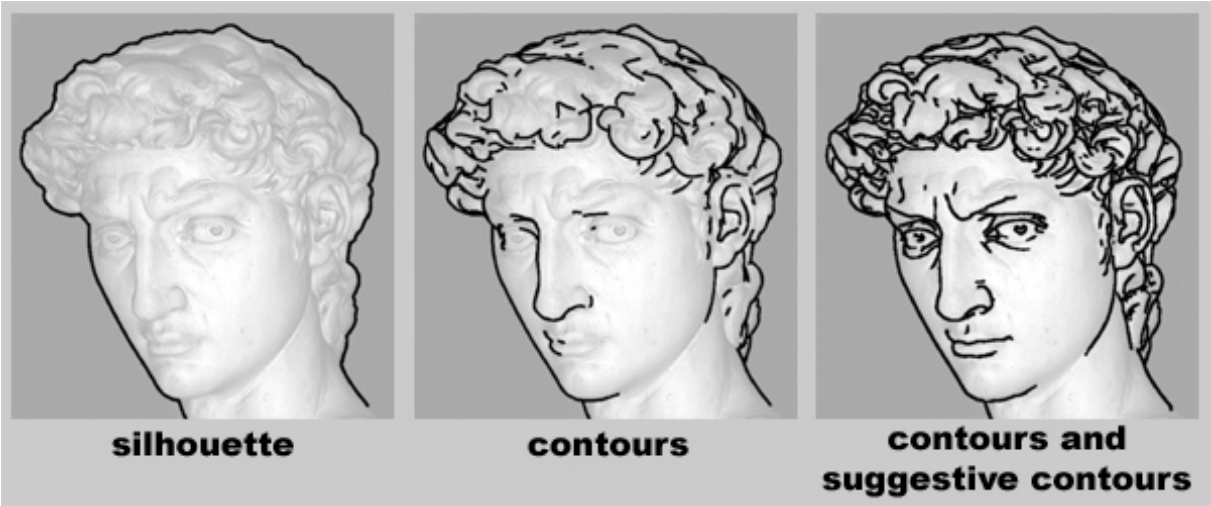


Figure 1.4. Mapping feature edges to lines. Image used with permission, © 2003 Doug DeCarlo.

Edges to Lines. My previous research examined automatic creation of technical illustrations from 3D geometric models by mapping view-dependent and view-independent feature edges to lines [**18, 17, 20**], as shown in Figure 1.3(a). DeCarlo *et al.* [**11**] generated additional lines in order to incorporate feature edges from nearby views, creating a mapping to lines from current edges and nearby view-dependent edges (Figure 1.4).

Lighting to Color. Spectral highlights and black edge lines often cannot be added to a fully shaded model because the highlights would be invisible in the light regions and the feature edges would be invisible in the dark or shadowed regions. To solve this problem, Gooch *et al.* [**18**] created a new shading model which varies light chrominance and luminance based upon lighting direction, making lit regions more yellowish and shadowed regions more bluish, creating a mapping from lighting direction to light color (Figure 1.3(b)).

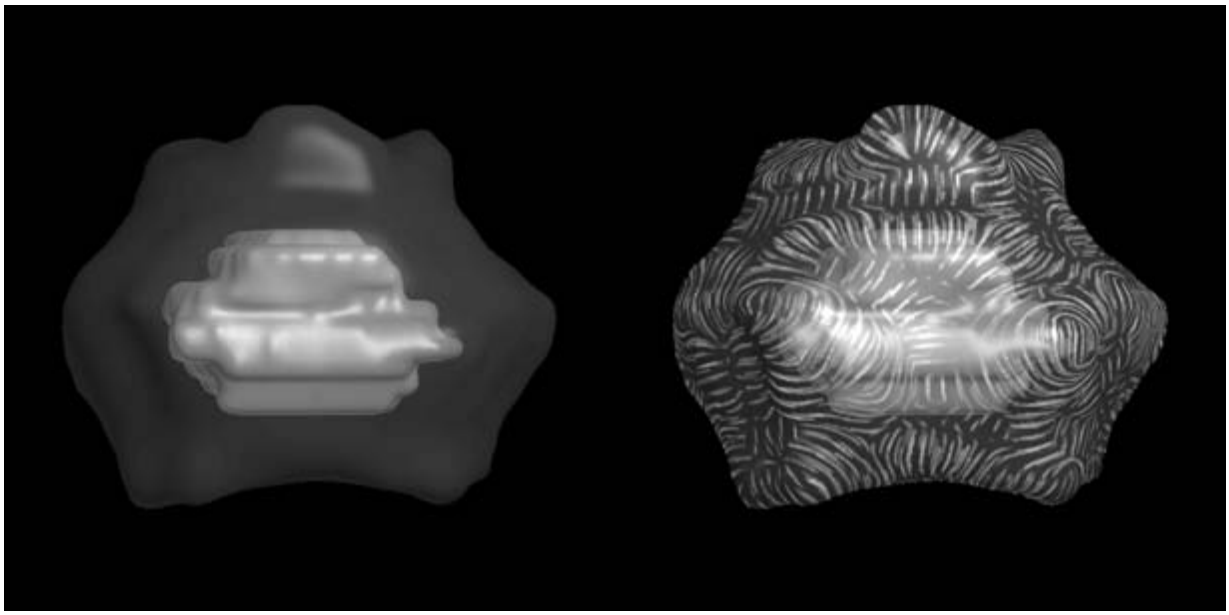


Figure 1.5. Mapping shape to texture. Image used with permission, © 1997 Victoria Interrante.

Shape to Texture. Research by Interrante and colleagues [**26, 25**] derived a mapping from shape to texture and performed user studies demonstrating that users can more reliably infer the 3D structure of the underlying shape when their shape-texture techniques are applied (Figure 1.5).

Abstract Quantities to Shape. Glyphs, or specific symbols that represent a semantic unit, are a common method for mapping shapes to abstract or concrete ideas. Healey *et al.* [**22**] rendered glyphs representing temperature (dark blue to bright pink for cold to hot), wind speed (low to high coverage for weak to strong), pressure (small to large for low to high), and precipitation (upright to flat for light to heavy) as shown in Figure 1.6. Such methods create a mapping from abstract quantities to shape and color, enabling the communication of complex, multi-dimensional ideas in a single image.

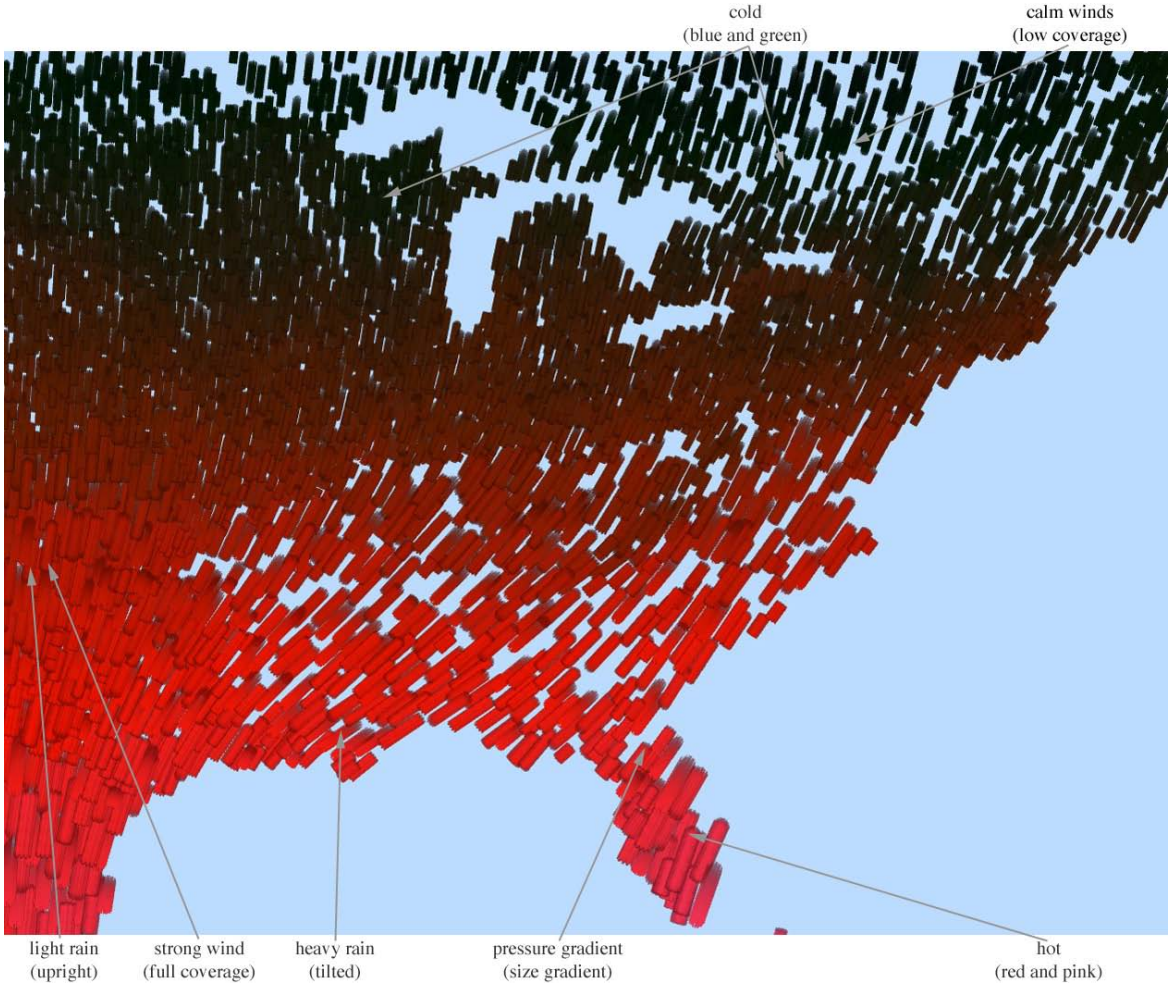
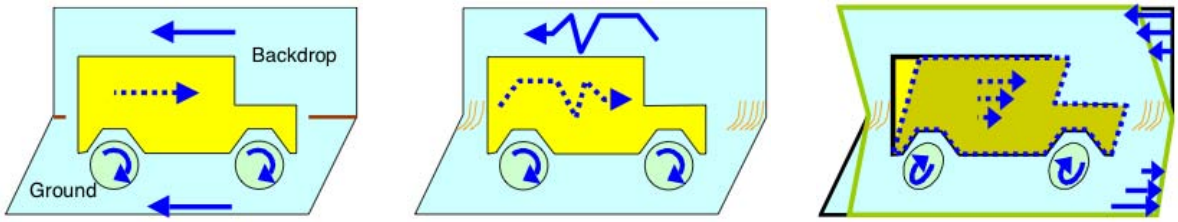


Figure 1.6. Mapping multi-dimensional quantities to shape. Healey *et al.* [**22**] rendered gylphs representing temperature (dark blue to bright pink for cold to hot), wind speed (low to high coverage for weak to strong), pressure (small to large for low to high), and precipitation (upright to flat for light to heavy). Image used with permission, © 2004 Christopher Healey.

Motion to Lines or Blurring. In cartooning, motion lines map changes in motion to visible lines in order to indicate that a person or object is moving quickly. As shown in Figure 1.7, motion lines convey a complex path of movement with a

(a) Motion blur with strobing (b) Strobing with motion lines

Figure 1.7. Mapping changes in motion to blurring or to lines. Image used with permission, © 2006 Aseem Agarwala and Antoine McNamara.



(a) Perceived horizontal forward motion, background texture moves backward, wheels rotate.

(b) Bumpy movement on a rough surface.

(c) Shear during acceleration.

Figure 1.8. Mapping changes in texture to motion. Raskar *et al.* [**43**] created apparent motion by changing textures on car body and wheels relative to the background (dashed lines indicated resultant apparent motion). Image used with permission, © 2002 Ramesh Raskar.

few simple lines in a static image. Motion blurring can also indicate relative motion in a static image.

Texture to Motion. Using images projected onto 3D shapes, Raskar *et al.* [**43**] created a cartoon-like environment in which a static 3D model of a car appears to move. Their method induced the appearance of motion using shadows, lighting, and texture, forcing temporary 3D interpretation errors. By changing the textures

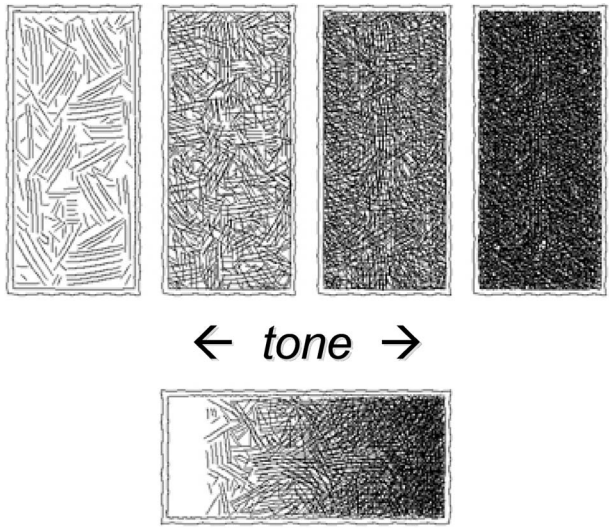


Figure 1.9. Mapping luminance to line density. Computer-generated pen-and-ink illustrations map illumination changes to texture or line density changes. Image used with permission, © 1994 Georges Winkenbach.

on the wheels and body of the car, instead of changing the backdrop, they were able to alter the apparent motion of the car moving forward, on bumpy surfaces, or even shearing to indicate sudden acceleration (Figure 1.8).

Luminance to Line Density. Early non-photorealistic rendering techniques for computer-generated pen-and-ink illustration [**51, 44**] mapped illumination changes to texture or line density changes (Figure 1.9) in order to visually indicate light direction. While line features alone cannot show shading, line density or spacing can convey surface tone and therefore can effectively replace smooth shading effects.

Multiple Depths of Field mapped to Focus. Agarwala *et al.* [**2**] created an image with an extended focal range from a series of image taken with different depths of focus, as shown in Figure 1.10. This mapping creates a single image with a large depth of field from multiple images with different focal lengths.

(a) Input 1 (b) Input 6 (c) Input 12 (d) Result

Figure 1.10. Mapping from images with different depths of field (Input 1 .. 12) to create a single image with an extended depth of field. Image used with permission, © 2004 Argawala *et al.* [**2**].

Time to Position. Line scan cameras capture the photo-finish of events, such as the race show in Figure 1.11, and create a mapping from time to position in the final image. This type of mapping records which body part crossed the finish line first; as can be scene with the seemingly impossible middle figure in Figure 1.11, the foot arrived at nearly the same time as the head, but stayed planted at the finish-line while the rest of the body proceeded forward. The line scan camera enables the capture and display of such finish line details in a single image.

Deviations in Shape to Exaggerations in Shape. Facial caricatures map deviations of features in a source face from an “average" face to exaggerations in the source face. Studies conducted by Gooch *et al.* [**19**] show that these exaggerations in black-and-white illustrations, as shown in Figure 1.12, are as effective as photographs in recognition tasks. Additionally, the mapping of grayscale photographs to black-and-white illustrations and caricatures also decreases the learning time for new names presented with new faces.

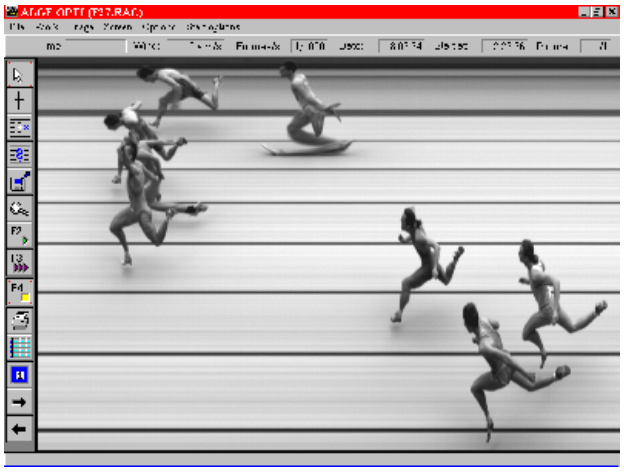
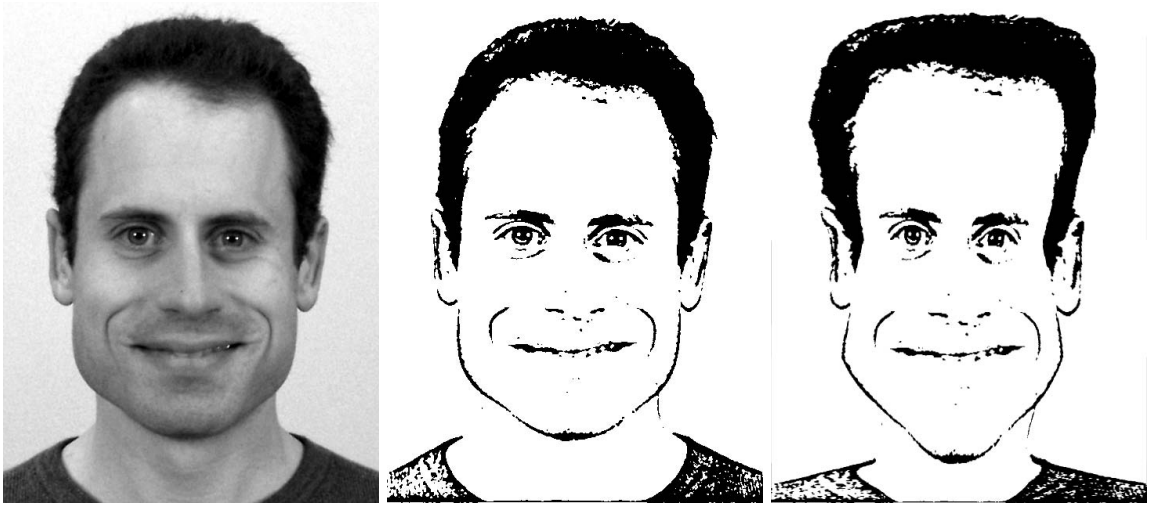


Figure 1.11. Mapping time to position. Line scan cameras capture the photo-finish of events such as the race show above and create a mapping from time to position in the final image. Image used with permission, ALGE Sports Timing Systems [**3**].



(a) Photograph (b) Illustration (c) Caricature

Figure 1.12. Caricatures map deviations in shape to exaggerations in shape. Image used with permission, © 2004 Bruce Gooch.

Position to Velocity. Blink comparisons map changes in position to changes or reversals in velocity. The American astronomer Clyde William Tombaugh discovered the planet Pluto in 1930 [**28**] by capturing photographs of sections of the

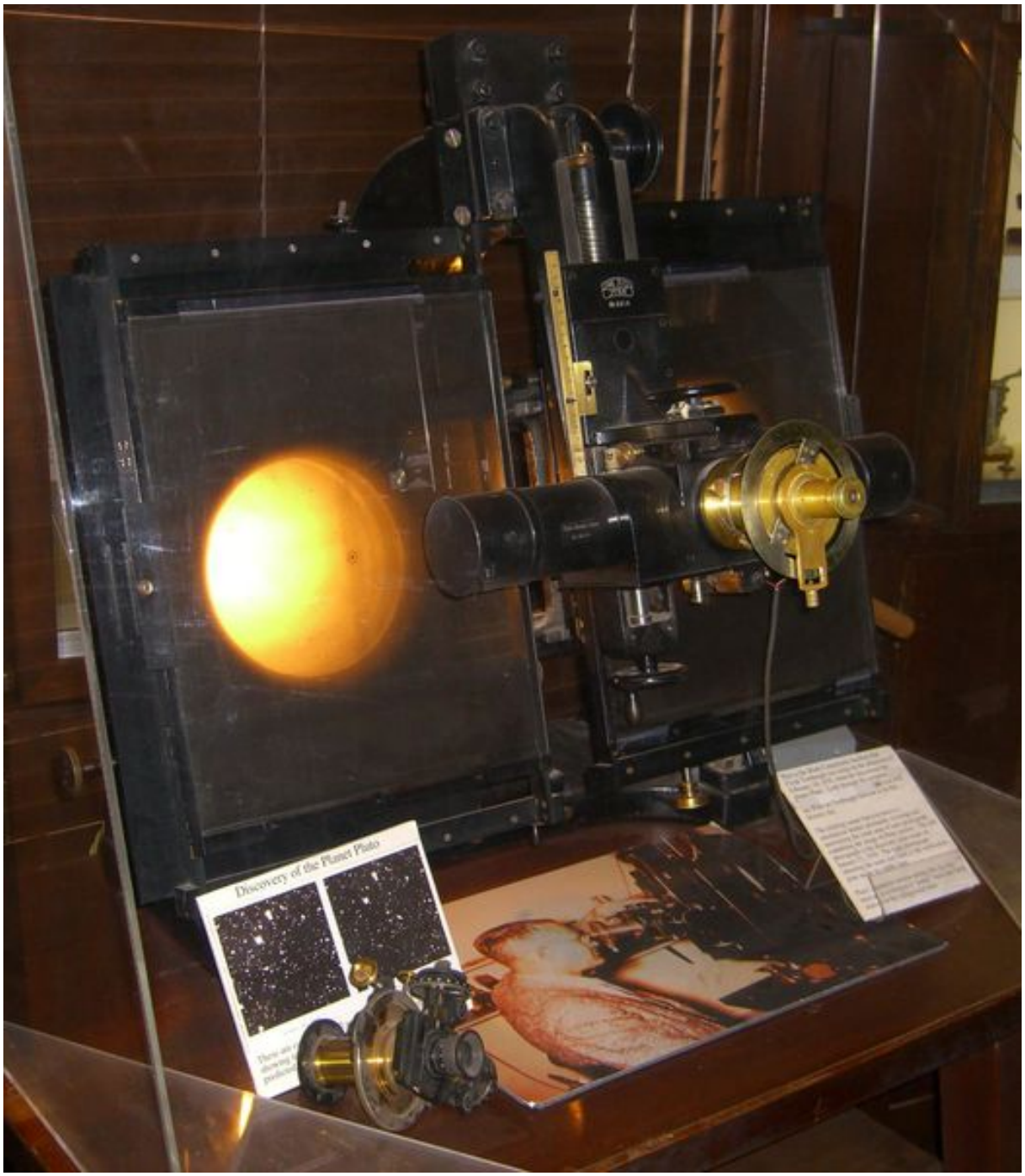
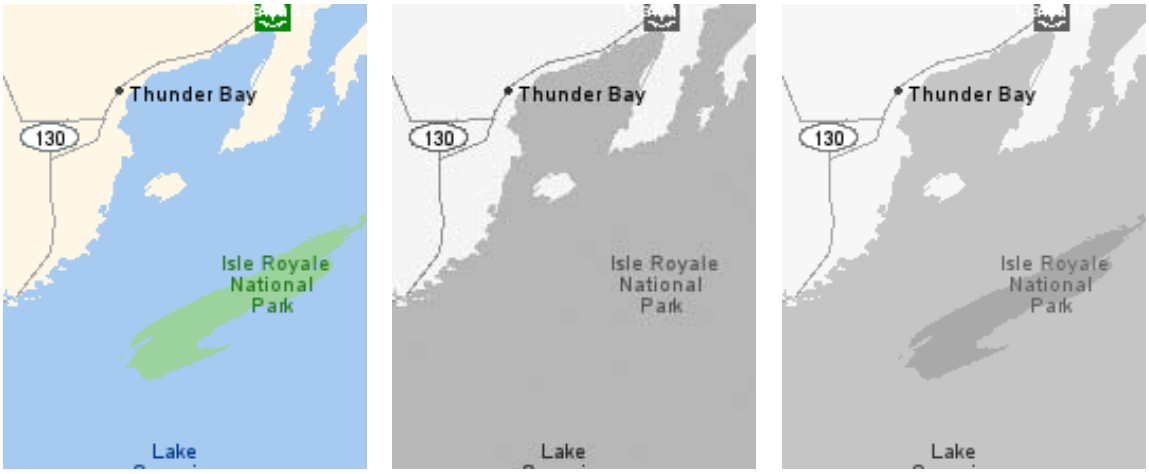


Figure 1.13. Mapping position to velocity. Blink comparator at Lowell Observatory used in the discovery of Pluto. Image from Wikipedia and permission to distribute is granted under GNU Free Document License.

sky taken several nights apart and then comparing them via a blink comparator, shown in Figure 1.13. The blink comparator enabled Tombaugh to compare the two images with the assumption that any moving object such as a planet would appear to jump from one position to another, while the more distant objects such as stars would remain stationary.

Although some off-axis mappings have been explored piecemeal, the matrix in Figure 1.2 reveals a much larger domain of maps that may hold valuable new image representations. Perhaps by methodical explorations of mappings from every domain to every other domain, many more new and important rendering possibilities may be found. For instance, mappings that preserve the magnitude of the difference between regions can yield images with a much larger number of visual cues due to the increase in freedom to map intensity and color values to a much larger set of parameters, such as texture, lines, or glyphs. More importantly, information in a scene can be conveyed by mapping scene region differences to differences, or changes, in nearly any renderable visual parameter. Considering image creation and modification as matrix of change-based mappings leads to several interesting research questions: instead of manipulating digital images as independent pixel values can I manipulate them in a way that preserves perceived changes? What does this type of image manipulation do to the semantic information in the image? My work contributes to the growing trend in computer graphics of using change-based mappings for image representation and manipulation.

Specifically, this dissertation will discuss two new mappings to address these two problems:



(a) Original Color Image (b) Default Grayscale (c) New Color2Grey

Figure 1.14. *Comparison of traditional and new color-to-grayscale conversion. a) Source color image. b) Converting the source image to grayscale by discarding all but the CIE Y channel results in the loss of important image features visible in the color image, such as the sun and its reflection in the water. c) Preliminary results of our algorithm incorporate changes available in the chrominance channels into a new luminance-only image. Image used with permission, © 2006 Yahoo!/NAVTEQ Maps.*

- **Color to Gray conversion.**

  Consider converting an image from color to grayscale, shown in Figure 1.14. Visible boundaries between isoluminant colors are lost in the default grayscale conversion. In Chapter 2 I will present a method to incorporate the perceived boundaries in the color image into the grayscale image.

- **Finding "Pentimenti".** Investigation of paintings using X-ray imagery allows conservators to explore the *pentimenti*, or hidden aspects, beneath the first surface of a painting. The X-ray image analysis of paintings can be difficult to interpret because an X-ray image represents relative density of

all the paint layers superimposed. In Chapter 3 I will present an approach that uses region-based statistics to identifying visually significant differences between X-ray images and color photographs. These tools, developed for ongoing conservation projects at the Art Institute of Chicago, aid conservators in the exploration of the artist's creative decisions that led to a masterpiece.

The intellectual merit of this research is the creation of images specifically targeted for the human visual system, unlike previous research efforts that have focused on the mapping of images to monitors, projectors, or paper. The creation of change-based mappings for images may have a broad impact on the field of image representation and manipulation.

CHAPTER 2

Mapping Color-to-Grayscale

Color documents printed in grayscale are often indecipherable. Figures and graphs in papers with saturated colors look good printed in color, but when printed in grayscale, a “red line" may appear to be the same shade of gray as a “green line". Grayscale mappings of color images that are constructed solely by approximating spectral uniformity are often woefully inadequate because isoluminant visual cues signaled only by chromatic differences are lost [**29**], such as the reflection of the sun in Figure 2.1. In this chapter, I will discuss a color-to-grayscale conversion that incorporates the perceived boundaries in the color image into the grayscale image in order to preserve semantic content and both global and local contrasts present in the original color image.

Figure 2.1. Mapping color-to-grayscale while selectively incorporating chrominance changes in addition to luminance changes. A color image (Left) often reveals important visual details missing from a luminance-only image (Middle). The Color2Gray algorithm (Right) maps visible color changes to grayscale changes. Image: Impressionist Sunrise by Claude Monet, courtesy of Artcyclopedia.com.

If digital images are regarded solely as an optical record, then a grayscale image only needs to record light intensities using a flat spectral response. Current color-to-grayscale conversions already meet this goal. However, as viewers, we often expect a more ambitious result: we want digital images to preserve a meaningful visual experience, even in grayscale. We are less concerned with the accuracy of light intensities and more concerned with the preservation of visual cues that help us detect the most meaningful and important, or *salient*, scene features. Accordingly, a black-and-white line drawing is sometimes far more expressive than a color photograph, and a garish cartoon-shaded rendering can often make important features, such as the shape, position, and reflectance of an object, more apparent.

Based upon observations by vision scientists with regard to the relative assessments of chrominance and luminance in human perception (Figure 1.1), we propose that preserving relationships between nearby pixels in an image is much more important than representing absolute pixel values. This research contributes to the growing trend in computer graphics of using change-based mappings for image representation and manipulation [**15, 39, 27**]. The technical contribution of this work is a color-to-grayscale image conversion algorithm based upon the human visual system's sensitivity to change. Additionally, this research provides a new signed chrominance distance calculation in the CIE $L^*a^*b^*$ chrominance plane. The Color2Gray algorithm creates images that maintain the salience of color images by mapping chrominance and luminance changes in a source image to changes in a grayscale image.

Figure 2.2. Isoluminant changes are not preserved with traditional color-to-grayscale conversion. Converting an image of a reddish square whose luminance matches that of the blue background (Left) to grayscale (Middle) results in a featureless gray image. The Color2Gray algorithm incorporates chrominance changes (Right).

Finding the most appropriate method to depict color differences in grayscale is also central to this research. For example, there are no luminance changes in the color image in Figure 2.2; adding them might seem at first to be a distortion of the original grayscale values. However, I claim that all visible color changes should cause visible changes if the grayscale image is to convey a more effective visual experience in comparison to current methods.

2.1. Related Work

Previous methods for converting RGB images to grayscale employed dot products or weighted sums to map a three dimensional color space to a single dimension, as shown in Figure 2.3. Other methods adopted by programs like Adobe Photoshop [**1, 6, 49**], devised custom non-linear projections and required users to set image-dependent parameters by trial and error. These methods applied a fixed function to map a set of 2D manifolds in color space to a set of 1D points

(a) CIECAM97 Lum (b) $L^*a^*b^*$ Lum. (c) XYZ Lum.

(d) YCrCb Lum. (e) Auto Contrast (f) Color2Gray

Figure 2.3. Comparison of grayscale conversions by previous methods (a-d), Photoshop's grayscale mode with auto contrast (e), and the Color2Gray algorithm (f).

(luminance) and were ineffective at preserving chrominance differences between isoluminant pixels (Figure 2.4). Contrast enhancement techniques increased the dynamic range of the image, but still map isoluminant pixels with different chrominance to the same gray value (Figure 2.3e).

Converting a color image to grayscale is a dimensionality reduction problem. In the course of my research, I have explored both linear and non-linear dimensionality reduction techniques, such as principal component analysis (PCA) and space-filling curves. PCA can be employed to compute an ellipsoid in color space that is a least-squares best fit for the cloud of points formed by all the image color values (Figure 2.5(a)). Color values in the image can then be projected on a luminance

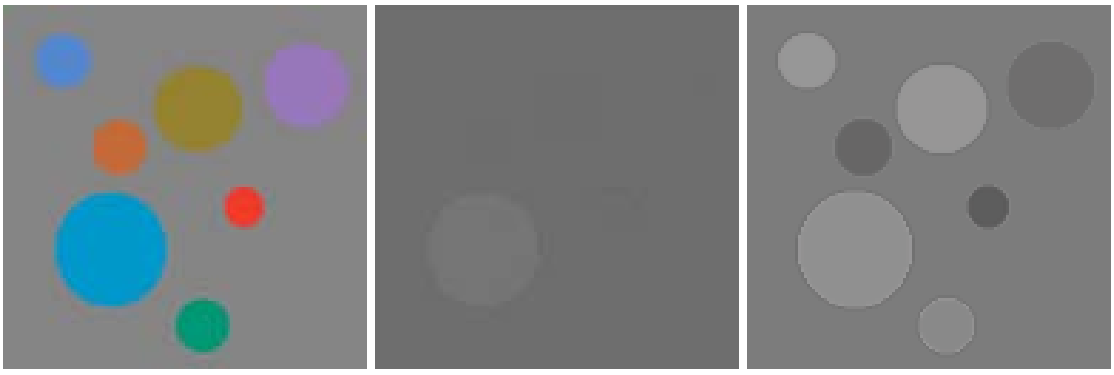
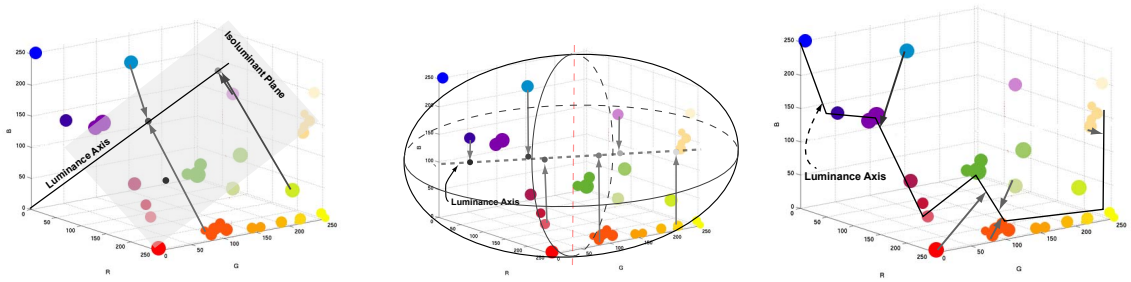


Figure 2.4. Converting the source image (Left) to grayscale with Photoshop's grayscale mode results in a nearly featureless image (Middle). The Color2Gray algorithm creates circles that are distinct from the background (Right). Parameters: $\theta = 180°$; $\alpha = 8$; $\mu =$ entire image.

axis defined by the primary axis of this ellipsoid. The effectiveness of PCA depended upon the color space; I found that the larger the ratio between the primary and secondary axes, the more likely the technique created distinctive gray values for different colors. For example, color space plots for a CIE $L^*a^*b^*$image appeared twisted and stretched in RGB space and yielded different principal component axes and different grayscale mappings. I also explored a non-linear dimensionality reduction technique, similar to Teschioni *et al.* [**47**], which created clusters in the color space and applied space-filling curves to build a 1D parameterization of 3D color space, illustrated in Figure 2.5(c). Both space-filling curves and PCA often created results with a high dynamic range because the length of any 1D pathway that adequately fills a 2D plane or 3D volume quickly approaches infinity.

I also experimented with Poisson solvers [**15**] and discovered that they work well on images such as the sunrise image (Figure 2.1), but not on images with



(a) Linear Mapping (b) PCA Mapping (c) Non-linear Mapping

Figure 2.5. Examples of several three-dimensional to one-dimensional mappings in color space. a) Traditional methods map values to an isoluminant plane; b) An example of a non-linear mapping; c) An example of a mapping computed with principal component analysis.

large disconnected isoluminant regions (Figure 2.4) because Poisson solvers compute gradients over nearest neighbors, ignoring difference comparisons over distances greater than one pixel.

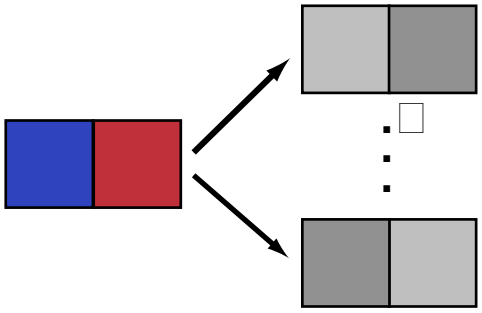
The color-to-gray problem is also similar to color quantization [**23**] and compression of gamut dimensionality. The research by Power *et al.* [**40**] created an image with a reduced set of inks or colors. However, both color quantization and gamut compression typically preserved luminance contrast and ignored contributions from chrominance for single ink/color images.

Contemporaneous research by Rasche *et al.* [**41, 42**] on color-to-grayscale conversion maintained the proportionality between perceived color difference and perceived luminance difference and ignored spatial arrangement of pixels. They also provided an extension to reduce three-dimensional color space to a two-dimensional surface to render images as they would appear to color-deficient observers. My solution incorporates changes from luminance, chrominance, and distance combined

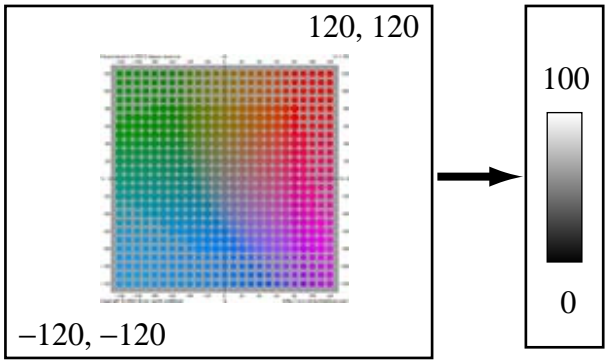
as well as provides three simple parameters to enable users to create aesthetic and perceptually salient grayscale images.

2.2. Challenges in converting to grayscale

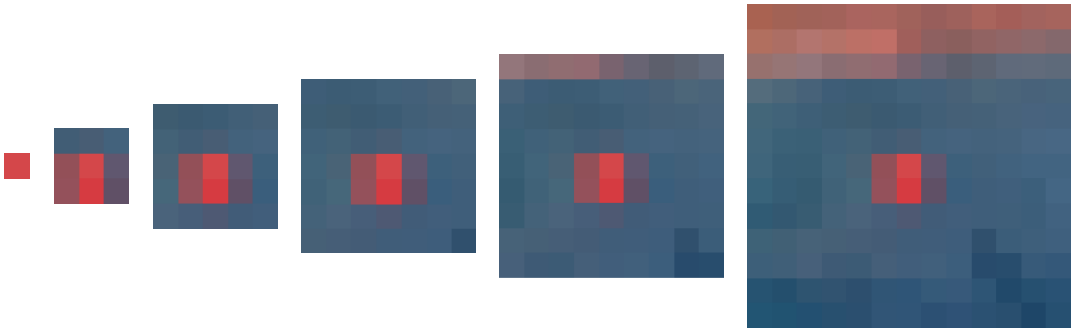
Generating a satisfying result sometimes requires making aesthetic decisions, such as the choice to make the orange sun brighter, not darker, than its isoluminant surrounding. Thus some level of user control is desirable. Previous interactive color-to-grayscale techniques required users to perform region-based adjustments, modify color channels or individually alter pixels. The Color2Gray algorithm treats the image as a whole, but encounters three main challenges when converting a color image to grayscale (Figure 2.6). The first challenge addresses whether to map color changes to lighter or darker luminance changes, such as was mentioned with adjusting the luminance to represent the sun on an isoluminant sky. The second challenge is to carefully map chrominance changes which can range in 2D from [-120,120] to [0, 100] without creating a fully saturated luminance image. The third challenge is to include the appropriate influence of neighboring pixels, as motivated by Figure 1.1; a color-to-grayscale algorithm must consider pixel neighbors larger than the eight surrounding pixels. These three challenges are controlled by the user in the Color2Gray algorithm via a three simple parameters (color-to-brightness angle θ and dynamic range mapping α, as well as neighborhood size μ), to be defined in Section 2.3.1.



(a) Challenge 1: Brighter or Darker



(b) Challenge 2: Dimension Reduction



(c) Challenge 3: Neighborhood Size

Figure 2.6. Three main challenges when converting a color image to grayscale.

2.3. Color2Gray Algorithm

The Color2Gray algorithm has three steps: first, the algorithm converts a color image to a perceptually uniform color space, then computes target differences in order to combine luminance and chrominance differences, and finally, uses a least squares optimization to selectively modulate the source luminance differences in order to reflect changes in the source image's chrominance.

The algorithm expresses the color differences between pixels in the color image as a set of signed scalar values, and then builds a grayscale version of the image with those values. For each pixel i and neighbor pixel j, a signed distance scalar, δ_{ij}, is calculated based upon luminance and chrominance differences between i and j. Using these values, a grayscale image g is derived. I refer to the grayscale difference between pixel i and a neighboring pixel j as $(g_i - g_j)$. A successful color-to-grayscale conversion consists of finding g such that all $(g_i - g_j)$ values closely match the corresponding δ_{ij} values.

Specifying δ_{ij} is fairly involved, and can benefit from minimal amounts of user interaction (Section 2.3.1). Once the δ_{ij} are formalized (Section 2.3.2), the algorithm finds the output image g by an iterative optimization process (Section 2.3.3).

2.3.1. Parameters

As discussed in Section 2.2, the Color2Gray algorithm allows users to control the mapping of color differences to grayscale differences via three simple parameters:

- θ: controls whether chromatic differences are mapped to increases or decreases in luminance value (Figure 2.7).

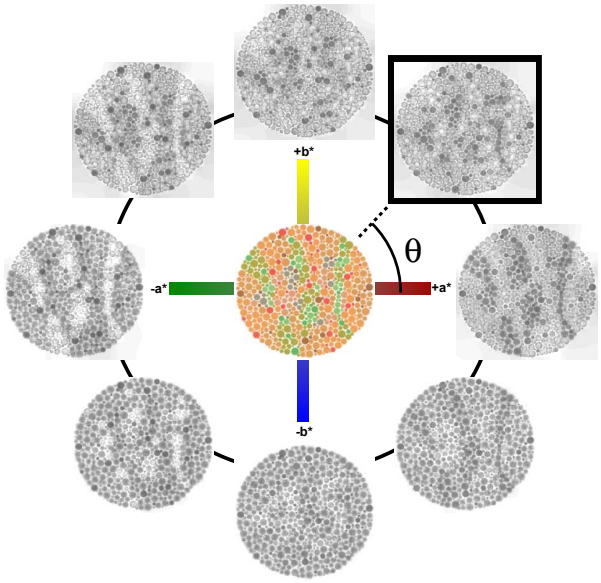


Figure 2.7. Color2Gray results for different values of the parameter θ, given the color image in the center of the figure. Note, for this colorblind image, you should not be able to see the “45" made of dots in the source color image unless you are color-deficient. Original image © 2006 Jay Neitz.

- α: determines how much chromatic variation is allowed to change the source luminance value (Figure 2.8).
- μ: sets the neighborhood size used for chrominance estimation and luminance gradients (Figures 2.9 and 2.10).

The angle θ divides the chrominance plane and determines whether a chromatic difference will darken or lighten the source luminance difference. To understand the intuition behind θ, consider mapping the chrominance changes in Figure 2.1 to shades of gray. The sun and its reflection would benefit aesthetically from mapping the oranges to brighter gray values (making the sun lighter) and mapping the blues to darker gray values. Such a cool to warm mapping corresponds to $\theta = 45°$.

Figure 2.8. Increasing alpha, α = 5, 10, 25 respectively, increases the contribution from chrominance differences but may cause dynamic range mapping problems ($\theta = 45°$).

Figure 2.9. **Left**: Equiluminant fade from gray to blue on a background with the same luminance. **Middle**: Color2Gray result with $\mu = 9$ neighborhood. **Right:** Color2Gray result with a full neighborhood ($\theta = 270°$; $\alpha = 8$.). Using a small neighborhood produces a dark band through the middle of the fade and produces artifacts where the edge of the fade meets the background; whereas using the entire set of pixels as the neighborhood (full neighborhood) preserves the horizontal fade with respect to its background.

Although I have implemented tools for automatically selecting θ, the parameter is available to the user. As shown by applying the Color2Gray algorithm to the color blindness test image of Figure 2.7, varying θ may provide insight into the salient cues available in the image. Appendix A provides several more examples.

The second parameter of the algorithm, α controls the amount of chromatic variation applied to the source luminance values. The luminance values, $L^* \in [0, 100]$, are in a much smaller range than either of the chrominance axes; a^* is in the range

[-500, 500] and b^* is in the range [-200, 200] [1]. The parameter α sets an upper and lower bound for how much of an effect a large chrominance difference can have on the current luminance value for a given pixel, such that the shift due to the chrominance values is within $[-\alpha, \alpha]$. In the next section, I define a function $\mathrm{crunch}()$ that allows us to compress large-amplitude values into the valid range, while leaving small yet significant variations around zero almost untouched. By default, $\alpha = 10$. However, some images may allow for more chrominance adjustment, such as the colorblindness test image in Figure 2.19, created with $\alpha = 15$. Figure 2.8 shows the effect of varying α.

The third parameter of the algorithm, μ, controls the size of the neighborhood, indicating whether a user is more interested in preserving local or global changes. Smaller neighborhoods may result in non-neighboring isoluminant regions with different chrominance values in the source image to be represented with the same grayscale value in the output image g. Additionally, preserving only local changes can create gradients that are not present in the source image, as shown in Figures 2.9 and 2.10. All of the figures in this document were generated using the entire image for neighborhood pixel comparisons, unless otherwise noted.

2.3.2. Computing Target Differences

The algorithm begins by converting the color input image to CIE $L^*a^*b^*$ color space, because its Euclidean distances (L2 norm) closely correspond to perceptual dissimilarity [**52**]. L_i refers to the luminance of the i^{th} pixel, and ΔL_{ij} is shorthand for

[1] Displayable chrominance values for most RGB monitors correspond more closely to a range of [-120, -120] to [120, 120]

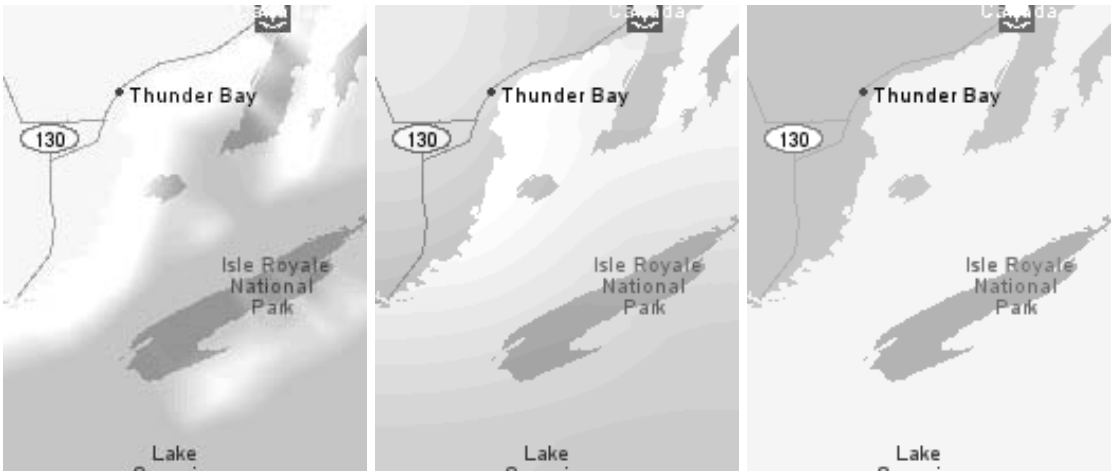
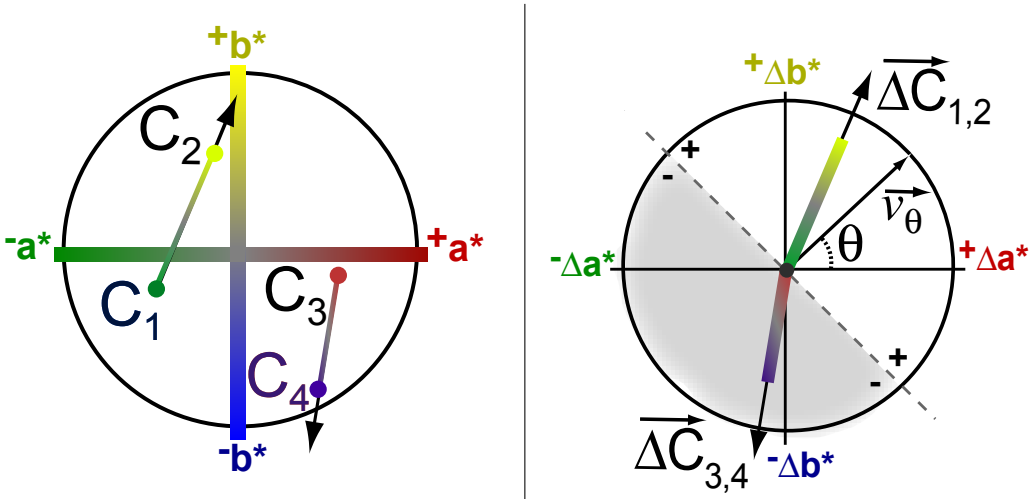


Figure 2.10. Comparison of results over neighborhood range: $\mu = 2$, $\mu = 16$, and the full neighborhood case, for an image with $\theta = 30$ and $\alpha = 10$.

$(L_i - L_j)$. Similarly, ΔA_{ij}, and ΔB_{ij} refer to the a^* and b^* channel differences between a pair of pixels. $\overrightarrow{\Delta C}_{ij}$ refers to $(\Delta A_{ij}, \Delta B_{ij})$, the chromatic difference vector between pixel i and it's neighbor pixel j.

In order to determine the target difference, δ_{ij}, I compare the luminance difference, ΔL_{ij}, with the chrominance difference, $\overrightarrow{\Delta C}_{ij}$. Since ΔL_{ij} is a scalar and $\overrightarrow{\Delta C}_{ij}$ is a 2D vector, I first map $\overrightarrow{\Delta C}_{ij}$ onto a single dimension using the Euclidean norm, $||\overrightarrow{\Delta C}_{ij}||$. Next I need to choose the appropriate sign for $||\overrightarrow{\Delta C}_{ij}||$ because it is always positive and δ_{ij} is a signed scalar. Therefore, I introduce an angle θ, which parameterizes the $\Delta a^* \Delta b^*$ chrominance plane. Let $\overrightarrow{v_\theta}$ be a normalized vector defined by θ relative to the Δa^* axis. I set the sign of the chromatic difference to be the same as the sign of $(\overrightarrow{\Delta C}_{ij} \cdot \overrightarrow{v_\theta})$, as illustrated in Figure 2.11. The parameter θ controls which chromatic differences are mapped to increases or decreases to the source luminance value (Figure 2.7), thus specifying the transformation of the $||\overrightarrow{\Delta C}_{ij}||$s into a signed quantity.



(a) Color Space

(b) Color Difference Space

Figure 2.11. Signed chrominance distance between pixels. Given (a) pixel pairs (i.e. $(C_i, C_j) = (C_1, C_2)$ or (C_3, C_4)), and (b) their chromatic difference, $\overrightarrow{\Delta C}_{ij}$, I divide the space of color differences into positive and negative halves based upon the parameter θ. I set the sign of the chromatic difference to be the same as the sign of $(\overrightarrow{\Delta C}_{ij} \cdot \overrightarrow{v_\theta})$, where $\overrightarrow{v_\theta}$ is a normalized vector defined by θ relative to the Δa^* axis, as illustrated in (b).

Finally, I define the target differences, δ_{ij}. If the absolute luminance difference is smaller than the chrominance difference, then I set δ_{ij} to be a measure of the chromatic differences; otherwise, δ_{ij} is set to the luminance differences. Formally, δ_{ij} is defined as follows.

Given: $\quad \text{crunch}(x) = \alpha * \tanh(x/\alpha)$

$\quad \overrightarrow{v_\theta} = (\cos\theta, \sin\theta)$

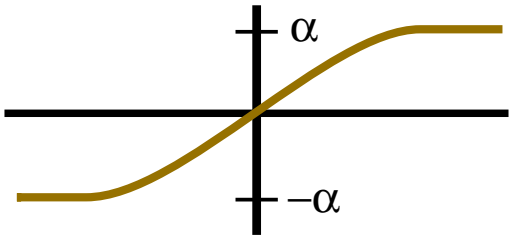


Figure 2.12. The function $\mathrm{crunch}(x)$, with parameter α, preserves small chrominance differences, while clamping large changes to $[-\alpha, \alpha]$.

then:

$$\delta(\alpha,\theta)_{ij} = \begin{cases} \Delta L_{ij} & \text{if } |\Delta L_{ij}| > \mathrm{crunch}(||\overrightarrow{\Delta C}_{ij}||) \\ \mathrm{crunch}(||\overrightarrow{\Delta C}_{ij}||) & \text{if } \overrightarrow{\Delta C}_{ij} \cdot \overrightarrow{v_\theta} \geq 0 \\ \mathrm{crunch}(-||\overrightarrow{\Delta C}_{ij}||) & \text{otherwise.} \end{cases}$$

I tried several methods for combining ΔC_{ij} and ΔL_{ij} in order to define δ_{ij}, and of those that I considered, the discontinuous case above worked best. Using a linear or smooth blend between ΔC_{ij} and ΔL_{ij} can make the resulting images muddier because when the two terms have opposite signs, the differences cancel out.

I can automatically determine θ by examining the spread of the chrominance values of the source image and finding an axis through the a^*b^* plane that divides these chrominances with the largest linear separation [**21**]. If the images do not have a single best axis, such as images with a large range of chrominance values, I utilize a cool-warm parameterization with $\theta = 45°$. However, for some images

and certain choices of θ, there may be troublesome discontinuities in the calculated signed color differences which will push the optimization in nearly opposite directions. I am currently investigating adaptive color difference signing and unsigned methods that would avoid these difficulties. However, in practice, these discontinuities have not produced noticeable artifacts.

2.3.3. Solving the Optimization

Given a set of desired signed differences δ_{ij}, I find a gray image g that minimizes the following objective function, $f(g)$, where $\mathcal{K}$ is a set of ordered pixel pairs (i, j):

$$f(g) \quad = \sum_{(i,j)\in\mathcal{K}} ((g_i - g_j) - \delta_{ij})^2. \tag{2.1}$$

I initialize g to be the luminance channel of the source image, and then descend to a minimum using conjugate gradient iterations [**45**]. Equation 2.1 is convex, but has multiple global minima, as for any grayscale image g, an image with an identical optimization score can be created by shifting all pixel values up or down by a constant value ($f(g) = f(g + c)$). In order to choose a single solution from the infinite set of optimal g, I shift the g vector returned by the solver such that it becomes as close as possible to the source luminance values (where “closeness" is defined using sum squared differences).

2.4. Performance of brute force algorithm

The cost of setting up and solving the optimization problem is proportional to the size of $\mathcal{K}$. By default, the algorithm compares every pixel to every other pixel,

i.e. $\mathcal{K}$ contains all ordered pixel pairs. Given such a $\mathcal{K}$, the algorithm scales poorly. However, one can also use a reduced set, such as that implied by only considering pairs that occupy a $\mu \times \mu$ neighborhood. For a square SxS image, the cost of generating all pixel pairs will be $O(\mu^2 S^2)$, or $O(S^4)$ for the full neighborhood (complete) case. Using an Althon™ 64 3200+, computing images with full neighborhoods requires 12.7 seconds for a 100x100 image, 65.6 seconds for a 150x150 image, and 204.0 seconds for a 200x200 image, just as you would expect for a $O(S^4)$ algorithm.

However, these runtimes can be improved considerably using programmable graphics hardware to perform the pixel comparisons in parallel. In theory, the ability to process an SxS block of values simultaneously should increase the performance of the algorithm from $O(S^4)$ to $O(S^2)$. In practice, the performance of the GPU implementation is closer to $O(S^3)$, requiring 2.8 seconds to solve a 100x100 image, 9.7 seconds for a 150x150 image, and 25.7 seconds for a 200x200 image, using an NVIDIA Geforce™ GT6800 graphics card.

2.5. Improving the algorithm

For the complete case (full neighborhood), I borrow an idea from Rasche *et al.* [**42**], and use a quantized version of the image to accelerate the calculation of δ_{ij}, dramatically speeding up the Color2Gray algorithm at the cost of a little accuracy.

More precisely, if c is one of the quantized colors, C is the set of all indices of the source image that are assigned to that color, and s_i is the color of the i^{th} pixel

of the image, then:

$$\sum_{i \in C} \delta(s_i, s_j) \approx |C| \delta(s_i, c). \tag{2.2}$$

Because finding δ_{ij} is the dominant cost of solving the complete case (full neighborhood), and a good approximation to δ_{ij} can be found in $O(N|Q|)$, the bound on the cost of finding this approximate solution will reduce to the cost of the quantization algorithm, which is presumably $O(N)$ or greater for N pixels and Q quantization levels. A similar acceleration could be arranged for the incomplete case (reduced neighborhood size), but because solving the incomplete case is actually at least as expensive as the δ_{ij} calculation, the gain is negligible. A more attractive optimization for the incomplete case is a multi-resolution solver, as demonstrated recently by Mantiuk *et al.* [**36**], which could be implemented independently of the quantization routine.

2.6. Color2Gray Evaluation

Two of the algorithm's parameters, α and θ, control the more subjective aspects of the color-to-grayscale conversion, and are therefore not practical to evaluate objectively. However, neighborhood size (μ) and quantization (q) warrant investigation. I employ the Visual Difference Predictor (VDP) for Low and High Dynamic Range Images [**34, 33, 35**] to evaluate the the Color2Gray algorithm over the neighborhood and quantization parameters. The VDP provides a score that predicts the probability that a human observer is able to perceive differences for a pair of images.

An analysis of 24 images for which the Color2Gray algorithm made perceivable changes shows that the effect of the parameter neighborhood size ($\mu = \{2, 4, 8, 16, 64\}$) varies from image to image. However, variations in the parameter quantization ($q = \{16, 32, 64, 128\}$) produces a negligible change in most Color2Gray results. Comparisons were made using the VDP algorithm between images produced under these values with the same source image produced with a full neighborhood and no quantization. Figures 2.13, 2.14, 2.15 and 2.16 present the data in two forms: the percentage of pixels for which the VDP is greater than 75% and greater than 95%, as well as a color coded per-pixel probability, with green [0,50%), yellow [50,75%), red [75,95%), magenta [95,100%].

Figure 2.13 shows that neighborhood size can have a large effect on perceivable differences, but that the strength of this effect strongly depends upon the source image. Images 2 and 8 are nearly the same isoluminant colorwheel image, created under two different color spaces, CIE L*a*b* and Photoshop LAB, but their scores vary slightly with different neighborhood sizes. Figure 2.14 shows VDP pixel statistics for Image 13.

Figure 2.15 illustrates that quantization rarely produces visible differences. Image 12 and 13 shows a slight perceptible difference. Image 13 contains a high-gradient region in the bright spot behind the blocks that would be most susceptible to quantization artifacts. The Color2Gray implementation uses the quantization routine provided by Image Magick Tools [**24**]. A better quantization scheme may be necessary for images that are more sensitive to the quantization scheme, however the percentage of affected pixels for a $q = 128$ is near zero, as shown in Figure 2.16.

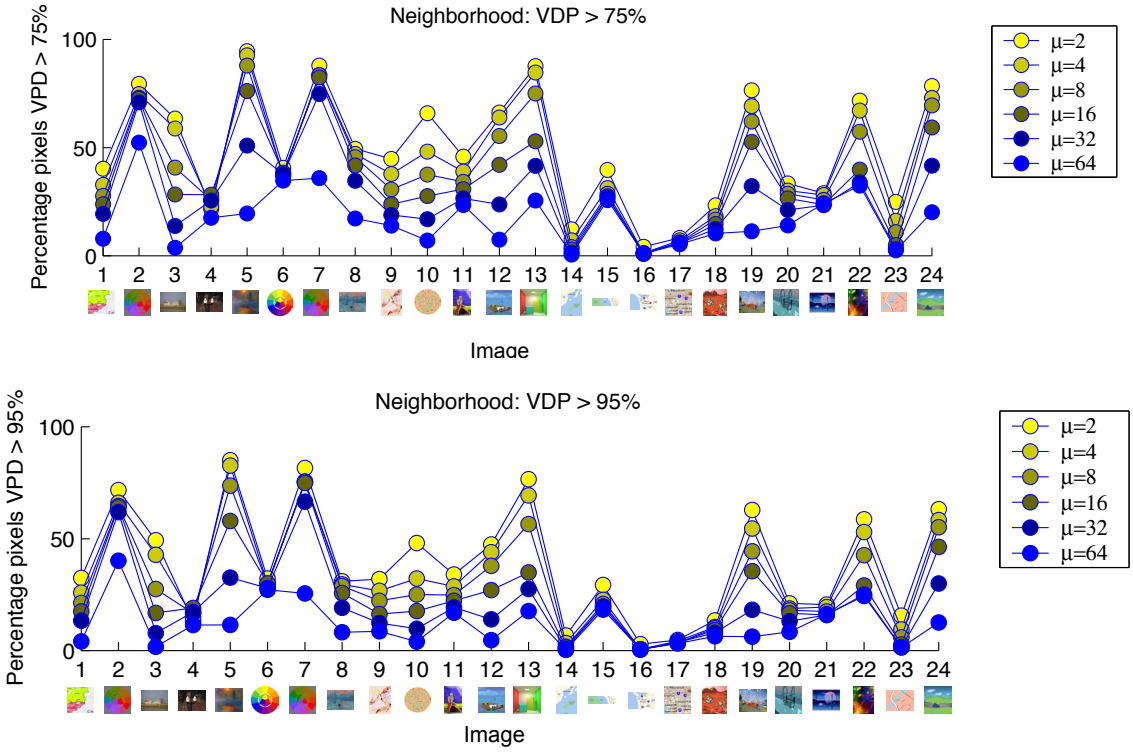
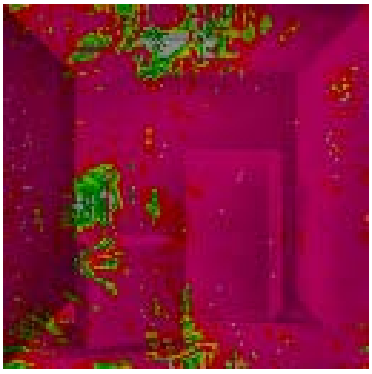


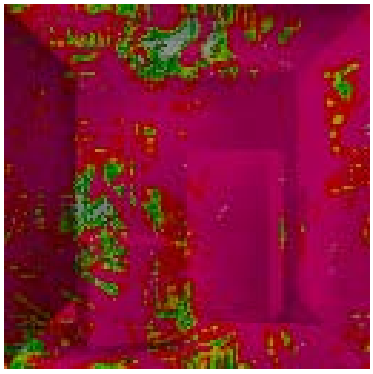
Figure 2.13. Comparing visible difference probabilities for images computed with full neighborhood size to images computed with neighborhood size (μ), for $\mu \in [2, 4, 8, 16, 32, 64]$. The graph illustrates that the data is highly dependent upon the input image (24 shown here, with inset of color original for illustrative purposes). VDP detector with 100% of pixels greater than a threshold (75% or 95%) indicates that all of the pixels in an target image have a high probability of being perceived as different from the source image.

2.7. Color2Gray Results and Discussion

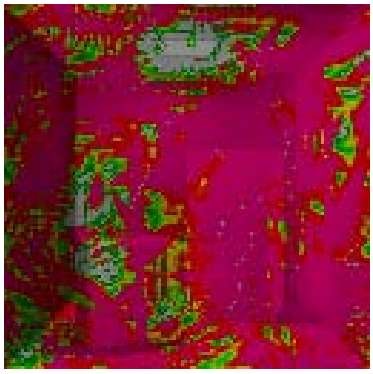
Figures 2.17, 2.18 and 2.19 compare the Color2Gray algorithm to Photoshop's grayscale mode on a wide variety of images. Isoluminant colors for state parks and water in the image in Figure 2.18, Row 2 make the island disappear in the Photoshop result. The effects of the Color2Gray algorithm are sometimes subtle,



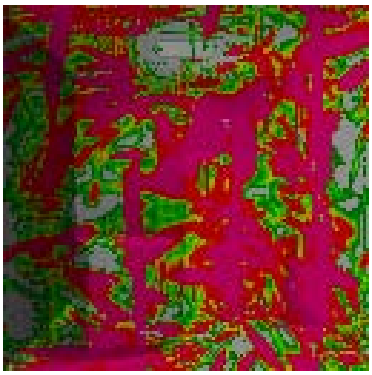
(a) Complete *vs.* $\mu = 2$



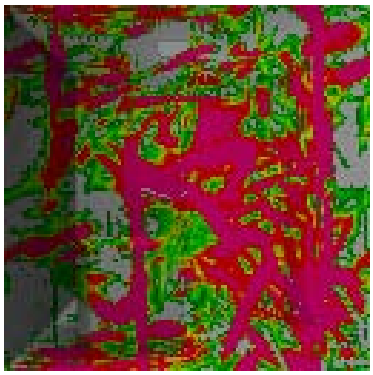
(b) Complete *vs.* $\mu = 4$



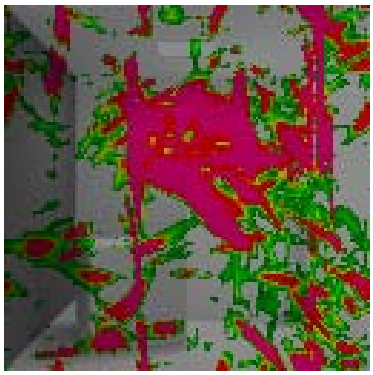
(c) Complete *vs.* $\mu = 8$



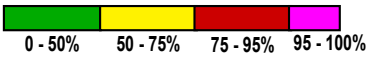
(d) Complete *vs.* $\mu = 16$



(e) Complete *vs.* $\mu = 32$



(f) Complete *vs.* $\mu = 64$



Compare complete case with	# pixels with VDP $> 75\%$	# pixels with VDP $> 95\%$
$\mu = 2$	87.71%	76.61%
$\mu = 4$	84.73%	69.33%
$\mu = 8$	75.04%	56.57%
$\mu = 16$	52.96%	34.91%
$\mu = 32$	41.60%	27.49%
$\mu = 64$	25.53%	17.64%

(g) Table of VDP probabilities averaged over all pixels

Figure 2.14. Smaller neighborhood sizes result in images that differ more from the full neighborhood (complete case). VDP probabilities per pixel and averaged over all pixels for Image 13 evaluated at four neighborhood sizes and compared to an image generated with full neighborhood size.

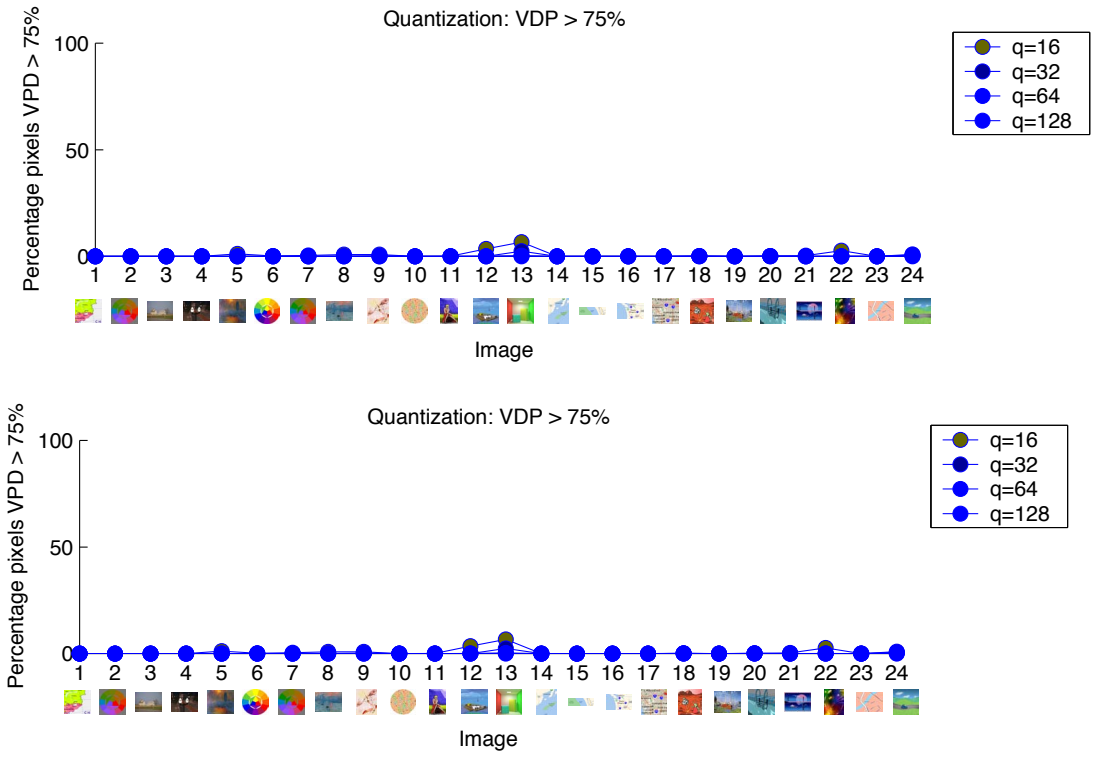
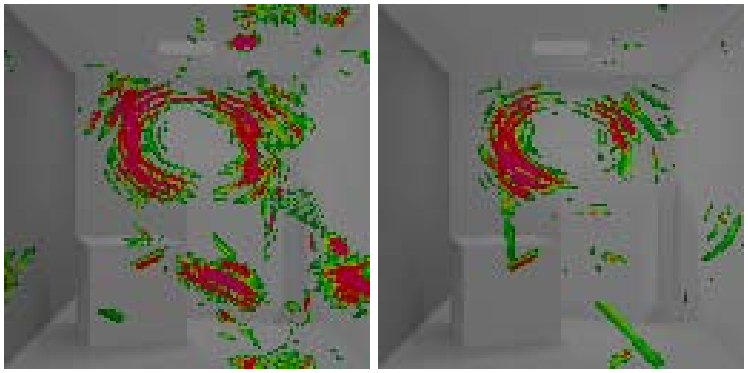


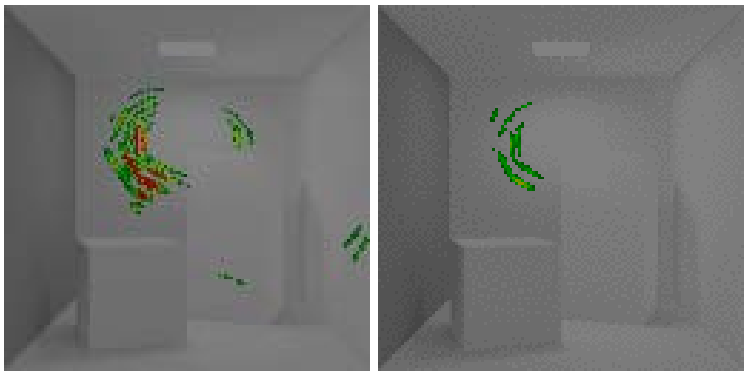
Figure 2.15. The graph of data gathered over 24 images, presented with inset of color original, illustrates that the quantization algorithm creates nearly no perceptible differences except in the case of Image 12 and 13. VDP detector with 100% of pixels greater than a threshold (75% or 95%) indicates that all of the pixels in an target image have a high probability of being perceived as different from the source image.

but color changes visibly darkened the region behind the head of the butterfly (Figure 2.18, Row 1) and darkened the green block while lightening the yellow block in the Cornell Box image (Figure 2.18, Row 4).

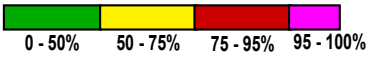
Our algorithm generally does not perform better than traditional grayscale representations when trying to map widely spread chrominance gradations to something independently distinguishable. The first row of Figure 2.20 illustrates this



(a) No Quant. *vs.* $q = 16$ (b) No Quant. *vs.* $q = 32$



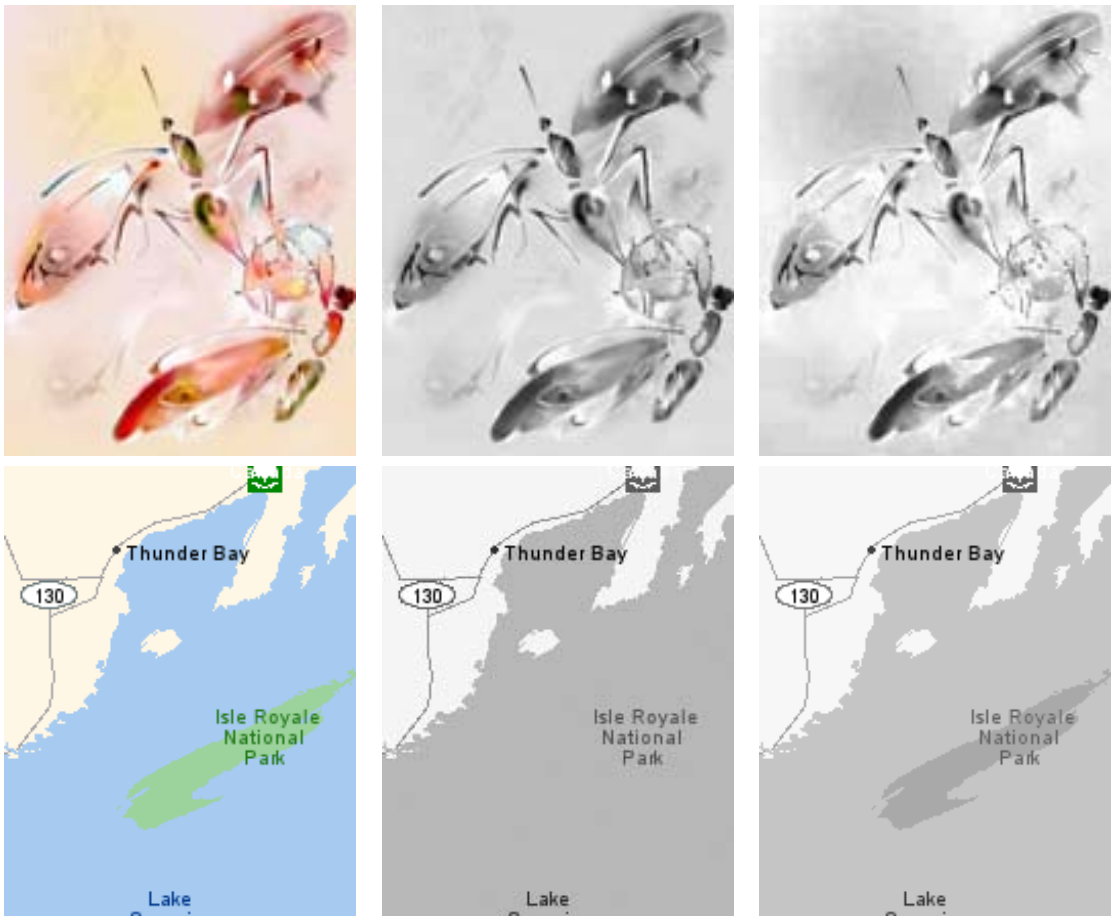
(c) No Quant. *vs.* $q = 64$ (d) No Quant. *vs.* $q = 128$



Compare no quantization with	# pixels with VDP $> 75\%$	# pixels with VDP $> 95\%$
$q = 16$	6.69%	3.36%
$q = 32$	2.42%	1.10%
$q = 64$	0.32%	0.01%
$q = 128$	0.00%	0.00%

(e) Table of VDP probabilities averaged over all pixels

Figure 2.16. Example images illustrating VDP evaluation for no quantization versus quantization values $q \in [16, 32, 64, 128]$. Although variations in the quantization parameter produce a negligible difference for most images, images with a high-gradient result in higher number of pixels with perceptible differences, especially for low values of q. This figure presents VDP probabilities per pixel and averaged over all pixels for Image 13 evaluated at four different quantization levels and compared to an image with no quantization.

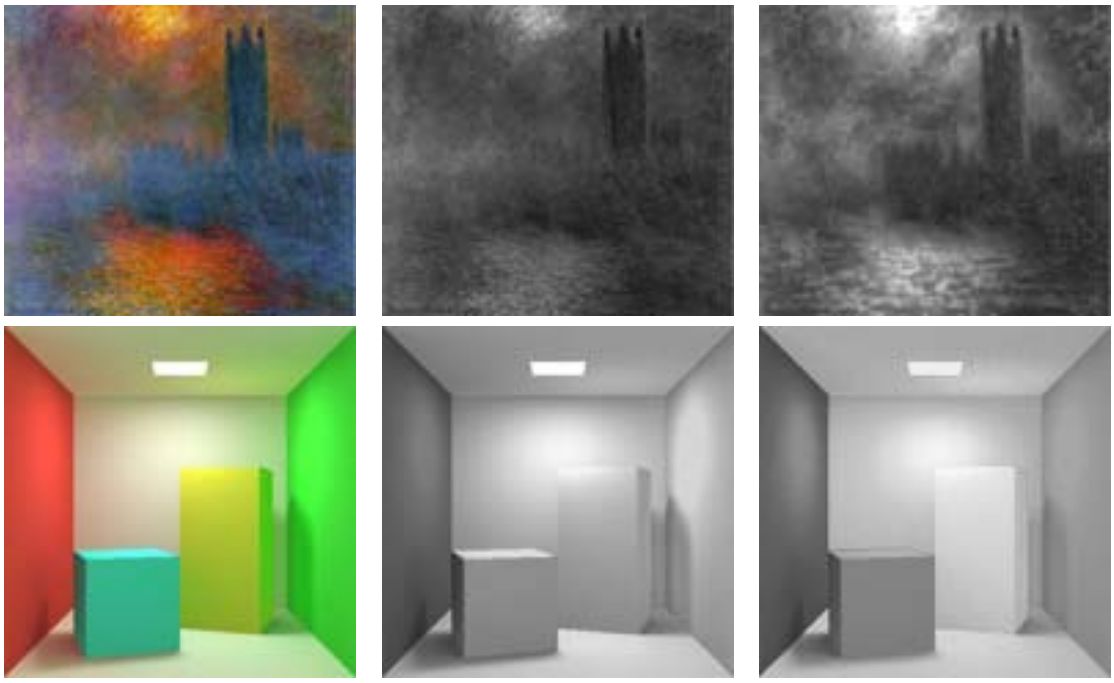


Color Source **Photoshop Gray** **Color2Gray**

Figure 2.17. Comparison of color source image, Photoshop grayscale, and Color2Gray results. Source images courtesy of Ana Vasileva, Yahoo!/NAVTEQ, respectively.

limitation. The image contains such a wide variety of isoluminant colors that no single span of 255 gray values can express them all separately.

The Color2Gray algorithm does not provide large improvements for scenes with high dynamic range, especially natural scenes, because they typically have a wide range of luminance changes. However, the method does improve any image that contains large isoluminant regions with a small number of different chrominance



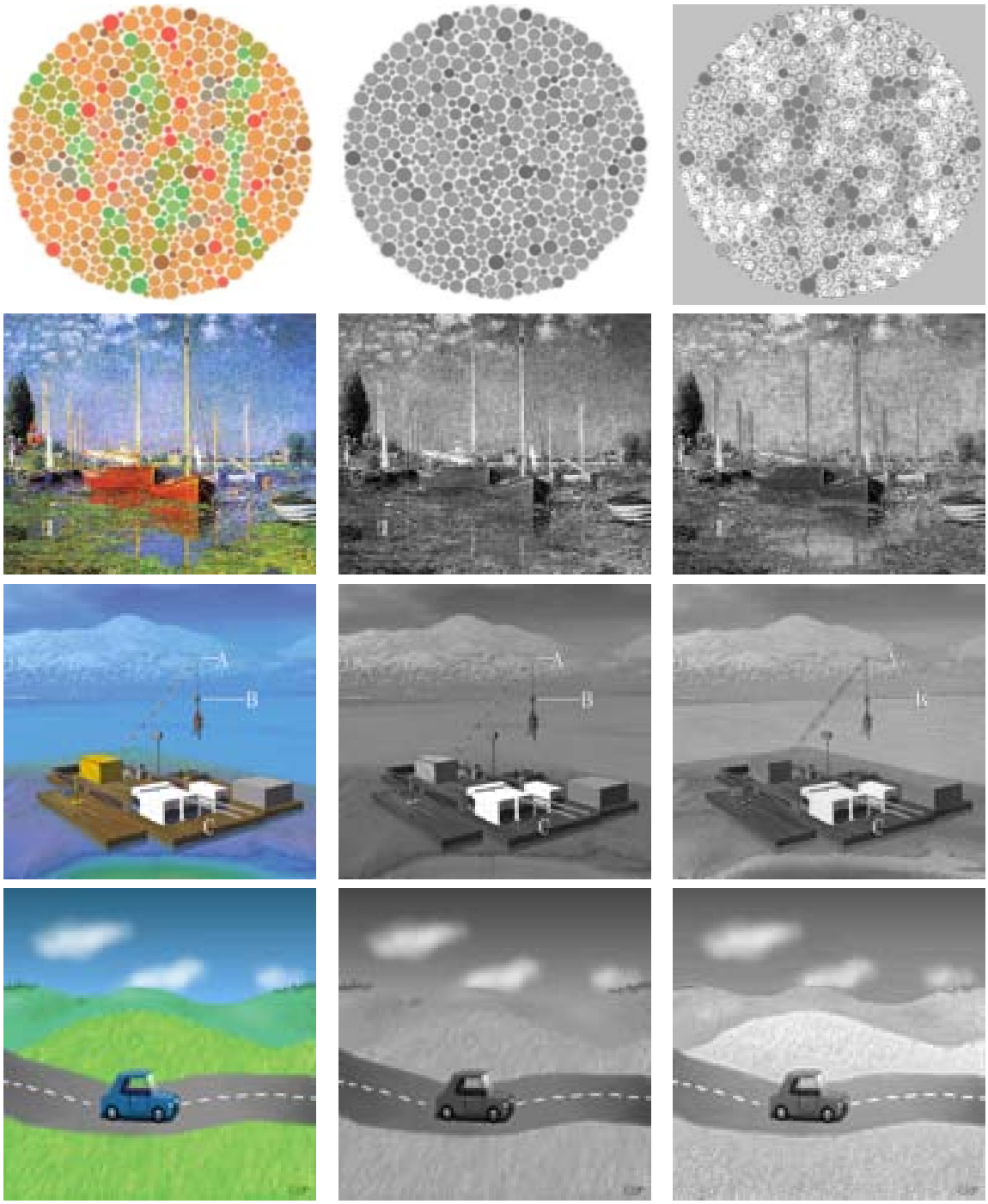
Color Source **Photoshop Gray** **Color2Gray**

Figure 2.18. Comparison of color source image, Photoshop grayscale, and Color2Gray results. Source images courtesy of Claude Monet, Dani Lischinski, respectively.

values such as the last row of Figure 2.19. Here, Color2Gray makes the two sets of hills behind the small car clearly distinguishable.

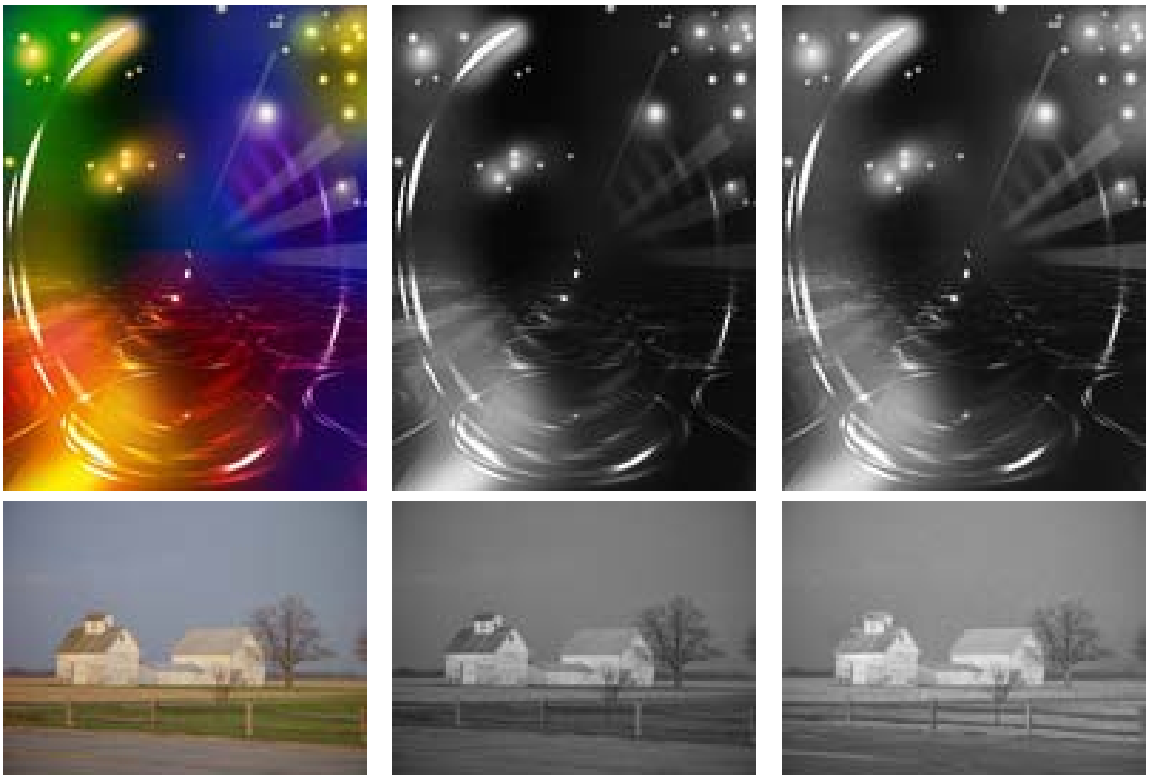
2.7.1. Augmented Color Images

I found that “Color2Gray result plus chrominance" images aid in making color images more easily comprehensible. I simply add the source image’s chrominance channels to the Color2Gray result in CIE $L^*a^*b^*$ color space and then convert the image to RGB. This technique modifies the color images only slightly but reveals Color2Gray results in a grayscale-only printout. By this method a document can be converted to one in which all of the figures will print legibly in both color and



Color Source **Photoshop Gray** **Color2Gray**

Figure 2.19. Comparison of color source image, Photoshop grayscale, and Color2Gray results. Source images courtesy of Jay Neitz, Claude Monet, Paul Chapman, and Leon Bli, respectively.



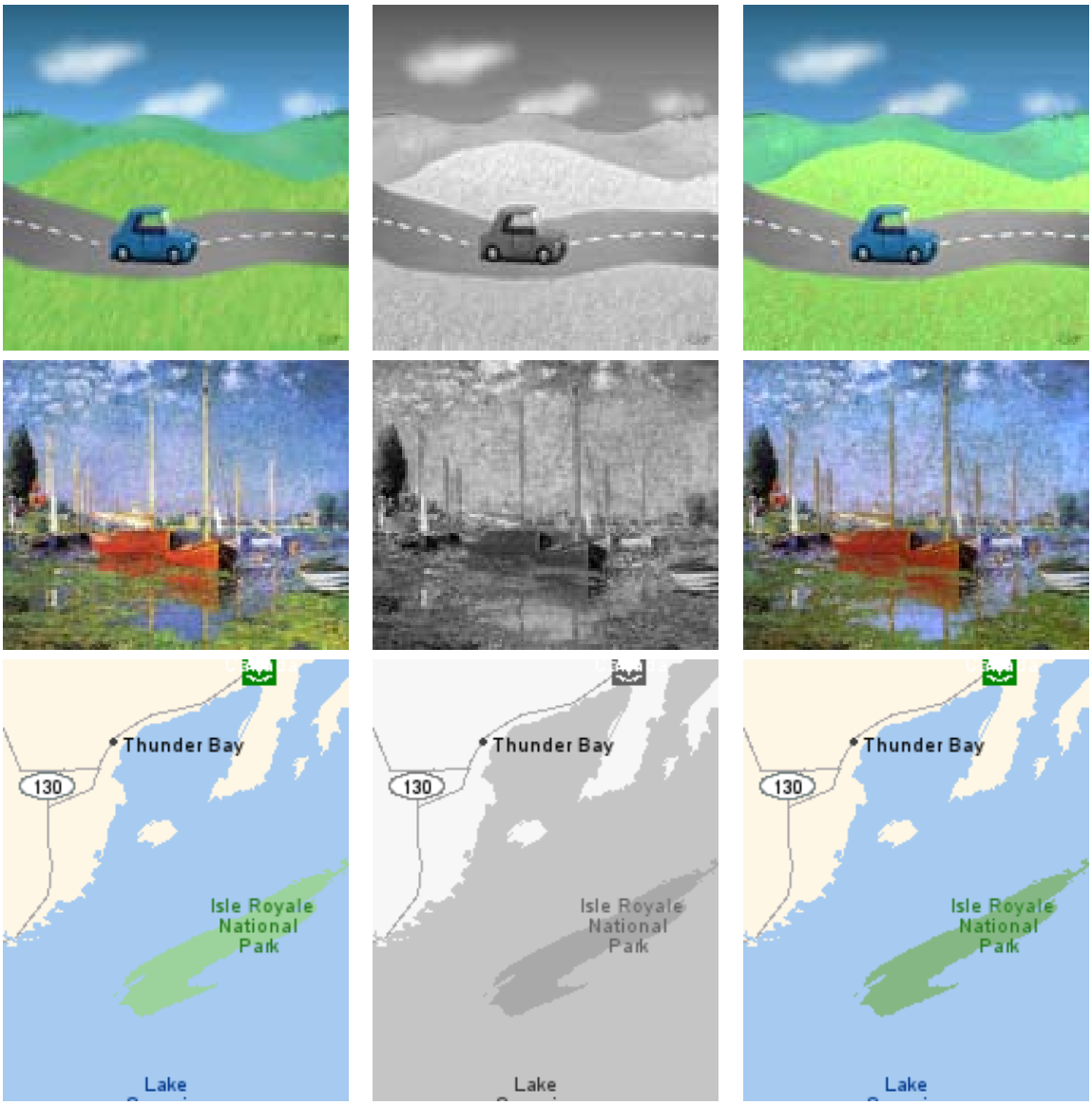
Color Source **Photoshop Gray** **Color2Gray**

Figure 2.20. Images with a wide range of colors (Top) or with subtle changes (Bottom) result in Color2Gray results that do not provide large improvements over Photoshop grayscale. Top source image courtesy of Ana Vasileva.

grayscale. Adding color to the Color2Gray result also provides useful feedback for visualizing the subtle adjustments made with the Color2Gray algorithm, as illustrated in Figure 2.21.

2.8. Future of Mapping Color-to-Grayscale

Extensions to the current Color2Gray algorithm could include removing the need to specify θ, for example, using an optimization function designed to match both signed and unsigned difference terms.



Color Source **Color2Gray** **Color2Gray +**

Figure 2.21. Color2gray results combined with chrominance (a^*b^*) from the source image make color difference more visible, and even grayscale printing reveals the improved Color2Gray result. Source images courtesy of Leon Bli, Claude Monet, and Yahoo!/NAVTEQ respectively.

Better target differences, δ_{ij}, could also include image complexity measures as well. For example, although a human can easily compare widely separated luminance and chrominance values in simple images (e.g. Figure 2.4, with colored circles on gray), comparisons in complex images over these same distances are far more difficult (e.g. Figure 2.1). In complex images, reuse of the same grayscale values for different nearby colors becomes more acceptable. Such investigations would require conducting perception-based experiments to understand the threshold for this difference as well as validating the degree to which our algorithm preserves the saliency of the color image.

2.9. Color2Gray Summary

The Color2Gray algorithm produces grayscale versions of color images, minimizing isoluminant mappings. While Color2Gray cannot improve all source images, I believe that its results will not be any less perceptually salient than the source luminance values alone. I encourage readers to print the images presented here on a high quality grayscale printer and examine their printed versions of the original color, Color2Gray, and Color2Gray enhanced images.

Additional example images can be found in Appendix A, and source code for the Color2Gray algorithm implementation can be found at `http://www.color2gray.info/`.

CHAPTER 3

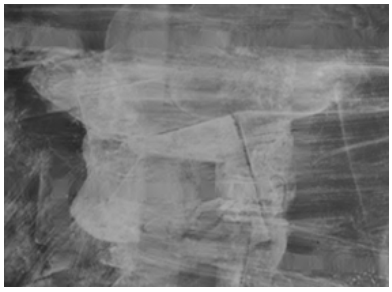
Mapping Difference in Visible Light and X-ray images

Art conservators often explore X-ray images of paintings to help find *pentimenti*, the artist's revisions hidden beneath the painting's visible first surface. X-ray interpretation is difficult because framing, canvas damage, and noise clutter the image. Additionally, X-rays superimpose features from all paint layers, and an X-ray image's intensity depends on both paint layer thickness and each pigment's opacity. I present a robust user-guided method to suppress clutter, find visually significant differences between X-ray images and color photographs, and visualize them together, as shown in Figure 3.1. These new techniques rely on region-based statistics in order to create a mapping of the regions that do and do not correspond in the visible light image and X-ray image of a painting. These tools, developed for ongoing conservation projects at the Art Institute of Chicago, allow domain experts as well as museum visitors to explore the artist's creative decisions that led to a masterpiece.

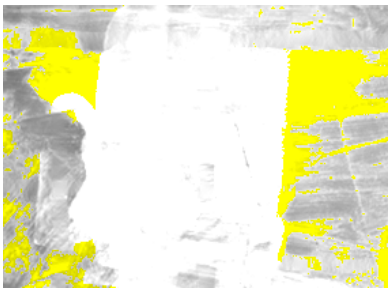
3.1. Visualizing Pentimenti: Exploring the Hidden History in a Painting

Occasionally the top layers of a painting wear away or become translucent to reveal the history of a painting. Since the Middle Ages, the art community has referred to the history hidden under layers of paint as *pentimenti*. Pentimenti, Italian for "regrets" or "repentance", refers to the changes undertaken in the course

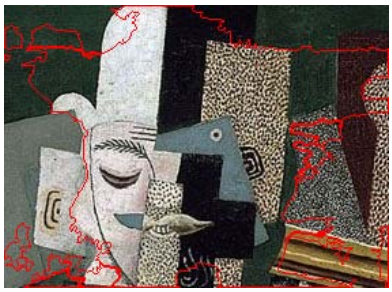
(a) Visible Image



(b) X-ray Image



(c) Mismatched Region Statistics



(d) X-ray Pentimenti

Figure 3.1. Visualizing the *pentimenti*, or hidden history of a painting, can aid art conservators in identifying regions in the first surface of a painting that are inconsistent with the paint density shown in an X-ray image. The user-guided visualization tool reveals a multiply revised hat in a portion of painting by Pablo Picasso, Untitled (Man with a Mustache, Buttoned Vest, and Pipe, Seated in an Armchair), 1915, oil on canvas, 130.2 x 89.5 cm, Gift of Mrs. Leigh B. Block in memory of Albert D. Lasker, 1952.1116 Reproduction, The Art Institute of Chicago. Visible light and X-ray images © Art Institute of Chicago, used with permission.

of painting a picture. Retracing the steps of a master artist can be immensely instructive, and help reveal artistic processes, decisions and influences that produced historically important paintings. Careful study and comparison of photographs of all sorts, from visible light images to the images gathered via X-ray, IR, UV, and fluoroscopic methods can sometimes reveal these steps.

This chapter reports upon a tool for exploratory data analysis that enables art conservators to explore and establish the significant differences between a visible light photograph of a painting and an X-ray image. An X-ray image records intensity values proportional to the amount the beam was impeded and reveals the deep structure of a painting and changes within layers because heavier atoms and thicker paint provide more resistance to X-ray transmission [**5**]. Although an X-ray image reveals a portion of the invisible history, it can be difficult to interpret because X-rays show all of the layers of the work superimposed. Conservation scientists, conservators, curators, and art historians, such as our colleagues at the Art Institute of Chicago, routinely obtain large quantities of X-ray data using a medical X-ray unit. However, the difficulty in interpreting these data lies in processing the multi-spectral data to remove artifacts and in finding and marking significant changes.

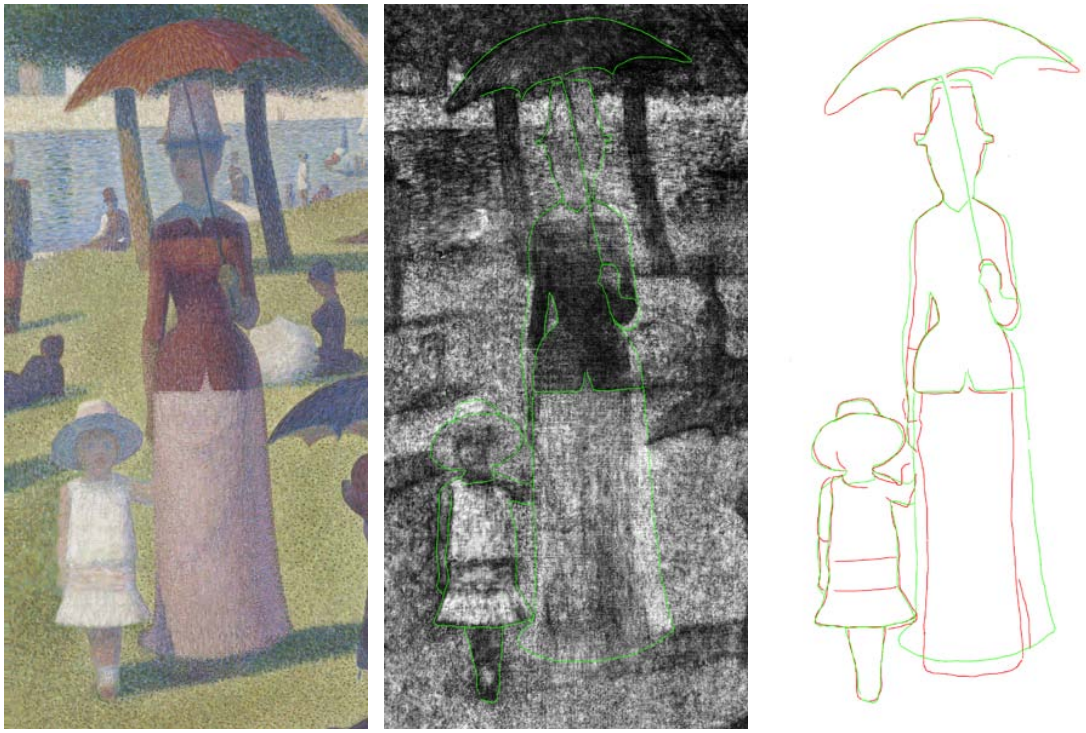
Through collaboration with the Art Institute of Chicago, I have access to a digital image archive of several paintings, including Pablo Picasso's "Half-length Female Nude", 1906, Autumn, 1959.619 and "Untitled (Man with Moustache, Buttoned Vest, and Pipe, Seated in an Armchair)", 1915, 1952.1116; as well as Georges Seurat's "A Sunday on La Grande Jatte – 1884", 1884-1886, 1926.224. The digital archive consists of the paintings as they appear on exhibit at the Art Institute of Chicago, as well as images of the painting captured through X-ray imaging.

Prior to our collaboration, the conservators at the Art Institute of Chicago combined an image of the painting and an X-ray of the painting into layers in Adobe

PhotoShop™ and used transparency to explore the differences. To present the differences between the X-ray image and the visible light photograph, conservators trace key outlines in both images and combine them over the X-ray or visible light image. Examining these layered images is a tedious process and it is difficult to manipulate multiple layers in one setting. An example image from the Art Institute of Chicago catalogue “Seurat and the Making of La Grande Jatte" [**48**] is shown in Figure 3.2. Our method helps draw attention to the differences between the visible light image and the X-ray image using false coloring and outlining. Users of the new system are only required to interactively manipulate simple parameters using a slide bar.

My goal is to aid conservators and other art experts in identifying the artist’s creative decisions by marking X-ray regions that differ from the final surface of the painting. The algorithm combines image statistics with mean shift segmentation to create final images that aid in the process currently used by conservators to compare X-ray images to visible light images. Through this investigation, I found that the new methods can aid but cannot replace an art conservator’s skills and judgment. The method documented here provides conservators and art historians with some tools that automatically mark significant features that differ between the X-ray and the visible light photograph.

Completely automated markings are difficult or impossible without semantic information. For example, although computers can easily outline a clump of bright pixels in the X-ray, they cannot easily delineate contextual information such as the



(a) Visible Light Image (b) X-ray Image (c) Both Lines

Figure 3.2. Hand-drawn analysis of sections of the painting before our visualization for “A Sunday Afternoon on the Island of La Grande Jatte" by Georges Seurat. a) Visible light image; b) X-ray image with overlaid green lines from final painting; d) comparison of X-ray contours (red) and visible light image (green); Images from painting by Georges Seurat, A Sunday on La Grande Jatte - 1884, 1884-86, oil on canvas, 207.5 x 308.1 cm, Helen Birch Bartlett Memorial Collection, 1926.224 combination of quadrant captures F1,F2, G1, G2. Reproduction, © The Art Institute of Chicago. Used with permission.

outline of a skirt that contains such marked pixels. With the exploratory data analysis tool, users can easily remove artifacts from the X-ray image and quickly identify regions in the X-ray that should be candidates for further exploration. Using this tool and the collaboration with conservators from the Art Intitute of Chicago,

we made several new discoveries, and leave the aesthetics of final presentation to the human hand.

3.2. Related Work

Researchers have examined several image-processing methods for the analysis and conservation of works of art [**4**]. ArtShop [**7**] provides image-processing tools for conservators to explore noise removal, color calibration, and virtual restoration for crack and lacuna filling (missing portions of the art work). Forth Photonics's "MuSis" hyperspectral imaging systems [**16**] provide hardware (CCD and tunable optical filter) and software to capture wavelengths from 365nm to 1.55μm (UV through visible to Near Infrared). Their software aids in the investigation of the spectral components of the captured images through grayscale images and histograms. They also compute false-color infrared images to combine visible and infrared light, in order to differentiate pigments having the same color but different composition.

Researchers at Hewlett-Packard worked closely with conservators to create a system for capturing the top surface of artworks illuminated from many different angles to uncover previously hidden details and explore the data set through polynomial texture mapping [**32**]. Most relevant to my research is a collaboration between imaging scientists at JPL and conservators [**12**] that used image processing techniques to clarify the radiography of the painting's lower layers. Their method

enhanced the contrast in the X-ray image, removed the periodic wood grain structure via a Fast Fourier Transform filter, and then applied gray-level matching, spatial filtering, subtracting, and averaging to produce an image of the under-painting. My research is in a similar vein, but allows a more active and controllable visualization, aiding in the discovery process.

Much of the research presented in this chapter relies on previous techniques in computer vision and image processing. There are several methods for evaluating the visual difference between two pairs of images such as Daly's Visual Difference Predictor (VDP) [**10**], and the High Dynamic Range Visual Difference Predictor (HDR-VDP) [**33**]. VDP methods examine images via amplitude compression, contrast sensitivity function filters, and spatial and orientation filters to produce probabilities of visible differences. However, a visible light photograph of a painting represents changes in color and luminance while an X-ray relates intensity to the relative density and thickness of paint. We cannot compare these two kinds of images in such a straightforward manner.

Additionally, researchers have investigated separating opaque foreground elements from background elements in an image in order to create alpha mattes and recompose images, such as Poisson matting [**46**] or Bayesian matting [**8**]. My problem is similar because I would like to separate the parts of the X-ray image that do and do not correspond to the visible light image. The Poisson matting approach required a user-defined "trimap" that denotes areas that are definitely background, definitely foreground, or an unknown mix of foreground and background. Iterative refinement using a Poisson solver collapses the unknown region to a refined,

line-like boundary. Bayesian matting [**8**] introduced the trimap and uses color statistics of the known foreground and background to estimate the unknown regions. Trimaps are problematic for most X-rays, because foreground and background elements appear transparent, overlapped, and are difficult to label consistently and unambiguously.

The method presented in this chapter applies robust mean shift segmentation [**9, 14**] to simplify input images into important regions of approximately uniform color or luminance. For each of these regions I compare statistical distributions of intensity between the visible light image and corresponding X-ray images. Inconsistencies between these two sets of distributions permit us to cleanly mark visually significant changes in the X-ray image.

3.3. Comparing Visible Light to X-rays

Figure 3.3 provides a pictorial view of each step in the method. First, the algorithm conditions the data to remove known artifacts, creating a new X-ray image; then the algorithm compares regions in the visible light image to intensity changes in the X-ray that indicate changes in paint density. Next, the algorithm automatically marks the inconsistent changes between a simplified visible light image and the new X-ray image. Finally, the algorithm gathers markings into regions and boundaries for annotation.

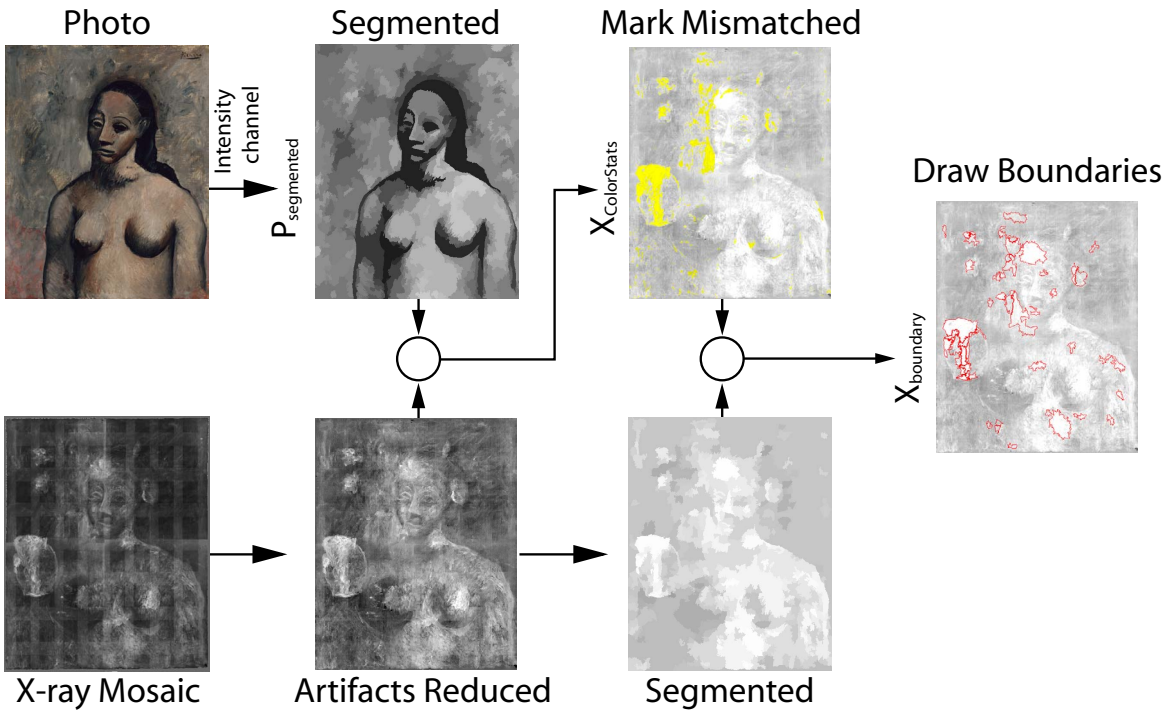


Figure 3.3. An overview of the process for finding visually significant features in the X-ray image that are not consistent with regions found in the first surface of the painting.

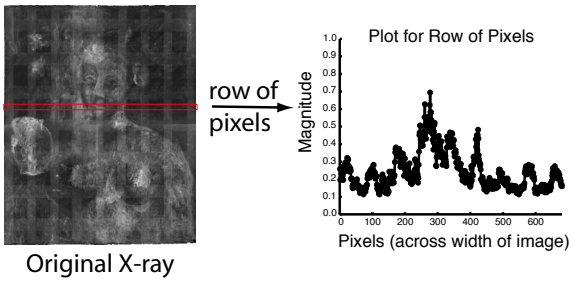
3.3.1. Conditioning the Input Data

X-ray image acquisition can be difficult for large paintings. Often conservators must stitch together many smaller X-ray images to compose a mosaic of the entire painting using standard image processing algorithms [**37**]. However, vignetting causes visible seams in the assembled mosaic. X-rays also collect many unwanted image artifacts, including shadows from painting support structures or cradles, canvas stretcher bars, braces, framing, nails, and even holes and trails left behind by canvas- and wood-eating beetles. As shown in Figure 3.4(b), artifacts often obscure much of the visually meaningful contents of an X-ray image.

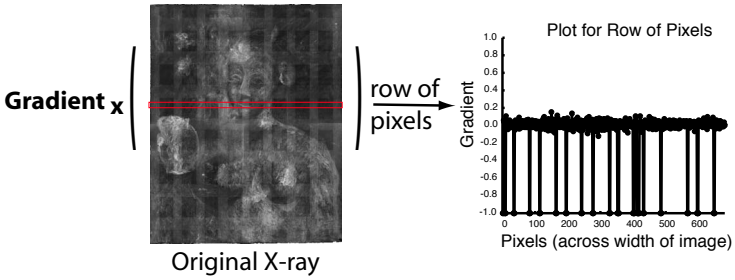
(a) Visible Light Image (b) Cradle behind painting (c) 8-image X-ray Mosaic

Figure 3.4. a) “Half-length Female Nude" by Pablo Picasso is painted on canvas attached to a cradled wood panel, shown in (b); The X-ray image (c) contains the properties of the density of paint in addition to the artifacts from the mounting structure. Painting by Pablo Picasso, Spanish, 1881-1973, Half-length Female Nude, 1906, Autumn, oil on canvas, 32 3/16 x 26 in. (81.8 x 66 cm), Gift of Florene May Schoenborn and Samuel A. Marx, 1959.619 Reproduction, The Art Institute of Chicago. Visible light and X-ray images © Art Institute of Chicago. Used with permission.

To remove these X-ray artifacts, I provide a user-assisted gradient editing program. Simply examining pixels alone does not aid in the identification of artifacts, especially those due to stretcher bars or other mounting structures captured in an X-ray. However, as shown in Figure 3.5, gradient-based analysis makes it easy to identify these changes in the X-ray. The algorithm calculates the forward gradients in the x and y directions and allows the user “paint" over the unwanted discontinuities. Everywhere the user “paints" with an $n \times m$ pixel brush (typically $2 \times m$ or $n \times 2$), the algorithm sets the gradient, which varies positively and negatively, to zero. Painting the x-direction and y-direction components separately allows the



(a) Pixel-based analysis



(b) Gradient-based analysis

Figure 3.5. Analyzing the pixel values alone will not aid in the identification of artifacts, such as the cradle board in this X-ray. Gradient-based approaches make it easy to separate these strong changes created by the cradle boards backing the Picasso painting shown in Figure 3.4.

identification of large horizontal and vertical bars without removing the underlying data. Then the algorithm uses a standard Poisson solver [**39**] to reassemble the gradients into a new luminance image for the X-ray image, as shown in Figures 3.6 and 3.8. The new pixel values and gradients, shown in Figure 3.7, preserve the significant features in the X-ray, features potentially hidden by the artifacts in the original X-ray.

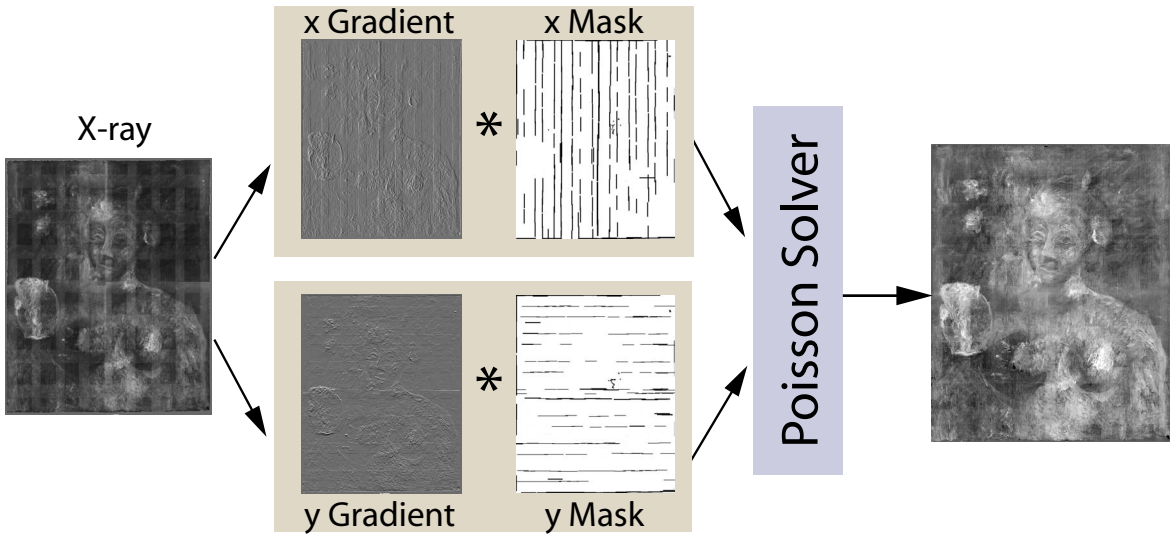
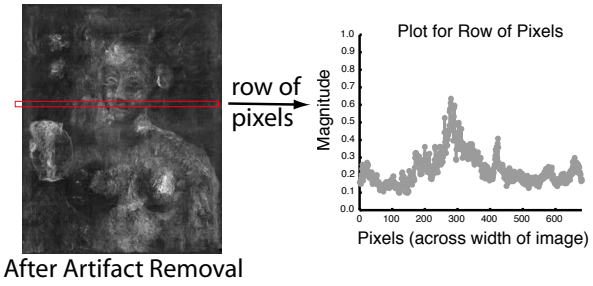


Figure 3.6. Editing the x and y gradients of the X-ray image allows us to remove cradle (or stretch bars) from the X-ray, as well as eliminate beetle marks, film flaws, and seams in mosaics of multiple X-ray images.

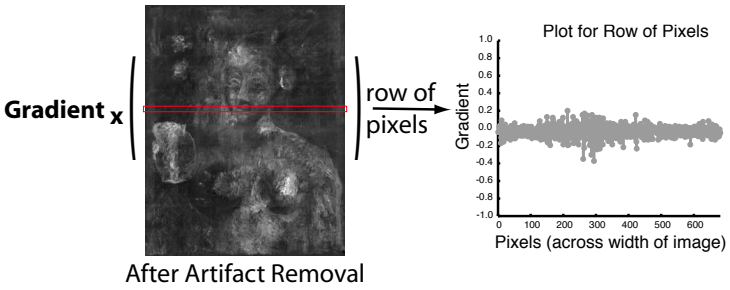
3.3.2. Detecting Mismatched Regions

After aligning the X-rays into an image mosaic, reducing as many artifacts as possible, and registering the visible light image with respect to the X-ray image, the algorithm compares the X-ray image to the visible light image. Intensity values in an X-ray image indicate the relative thickness and X-ray opacity of paint. However, the visible light photograph of a painting represents changes in color or luminance. Ideally, the algorithm should to find a mapping that allows the comparison of changes in thickness and density to changes in luminance or color.

In order to compare X-ray and visible light images, I assume a connection between the intensity changes in the luminance channel of the visible light image and the intensity changes in the X-ray that indicate paint density. Before the 1920s,



(a) Pixel profile after artifact removal



(b) Gradient profile after artifact removal

Figure 3.7. After gradient-based editing of the X-ray, we are able to obtain the important changes associated with the painting itself, and successfully remove several artifacts.

(a) X-ray (b) Edited X-ray

Figure 3.8. Editing the gradients of the X-ray (a) allow the removal of artifacts such as X-ray film seams and stretcher-bars, producing a clearer X-ray image (b).

most white pigments used in paintings contained lead, a very heavy element with high X-ray opacity. Therefore it is conceivable that the luminance channel of the visible light image conveys a distribution of the white pigment, and if the X-ray reflects the content of the first surface, then there should be a strong correlation between the luminance values in the first surface of the painting and the intensities of the X-ray image.

Therefore the exploratory visualization tool assumes:

- Visible light regions of approximately constant intensity should have approximately constant intensity in an X-ray image, but only if the paint thickness, paint material, and paint layers are uniform throughout the region.
- Inconsistencies in paint materials, thickness, and layers reliably indicate pentimenti; hidden revisions under the painting's visible first surface.

The above assumptions help with one possible problem with treating an X-ray image as directly dependent upon the density of the paint: different paints may impede X-rays differently due to elemental composition of pigments, such as the quantities of lead in the paint, as well as paint thickness. By treating each region in the painting differently, I do not directly need to account for different X-ray reflectance due to material composition of the different colors of paint, but admit I am only doing so with a coarse approximation.

My algorithm runs Mean Shift Segmentation [**9, 14**] in CIE L*u*v* color space on the luminance channel (L*) of the color image to create a simplified number of regions in a new segmented visible light image $P_{segmented}$, based upon parameters

at the users control, such as color radius (c=6) and spatial radius (d = 7), as well as the minimum number of pixels that constitute a region ($minRegion = 150$).

Then the algorithm separately considers each of these regions in the visible light and X-ray images. For each luminance region, the algorithm gathers the mean intensity and standard deviation from the mean on the corresponding intensity values in the X-ray image. Then, the algorithm produces an intermediate image, $X_{ColorStats}$, marking every pixel in the X-ray image that is above $x_{stdthresh}$ standard deviations from the mean. The marked yellow pixels, as shown in Figure 3.9, indicate regions where the intensity is $x_{stdthresh}$ standard deviations above the other paint densities associated with that the corresponding luminance region in the visible light photograph and indicate to the conservators that these are the areas most likely to be over-painted.

The previous steps allow the identification of visually significant intensity values in the X-ray with respect to large regions in the visible light image, creating a mapping of visually significant changes. The algorithm next creates local regions by examining the percentage of marked (yellow) pixels with respect to each region in a mean shift segmentation of the X-ray image (Figure 3.10). For each region, the algorithm draws the boundary if the percentage of marked pixels is above a user determined threshold, producing border regions in image $X_{boundary}$, as shown in Figure 3.10. Finally, the algorithm composites the boundaries of the important regions onto either the X-ray image or the visible light image for evaluation.

(a) $P_{segmented}$ (b) $X_{ColorStats}$

Figure 3.9. We approximate luminance regions in the visible light photograph through mean shift segmentation (c = 6, d = 7, $minRegion = 150$), shown in a); Yellow pixels in b) indicate X-ray intensities more than $x_{stdthresh} = 1.8$ standard deviations above the mean for the corresponding color region in a), indicating that these areas are most likely to be over-painted.

3.4. Visualizing Pentimenti: Discoveries and Discussion

Through the exploratory data analysis, and with the help Conservator Allison Langley, we found a few interesting features that were not seen with traditional methods, most notably in the “Half-length Female Nude" data set. Removing the cradle pattern in the X-ray image more clearly reveals the shape of the pitcher in the lower left of the painting. In addition, there is clearly a vase to the left of the figure’s face, next to her chin. This was hinted at in the original X-ray image, but was hard to read. The shape and placement of the vase and the oval opening at the top of the object is much clearer in the X-ray image after the clean-up process. It is possible that the artist used the edge of the dense white paint of the vase to place or position the left side of the figure’s face; the cheek, chin and shoulder follow the

line of this underlying form. Picasso left the white of the vase partially visible in the finished painting to contrast with the darkness of the face, which he painted later. The light highlights to the left and right of the chin bring the face and chin forward in space. Picasso often left portions of his underlying images visible to act as part of the upper compositions.

The cleaned up X-ray image also reveals the large, dark round platter or form that lies under the left breast, to the right of the pitcher, beneath the shoulder. This large round dark form is clearer in the new X-ray image. It is also faintly apparent in the visible light image of the painting, but more difficult to resolve. With the new images, it is easier to see that part of the underlying still-life included a pitcher, a vase, and a large round platter, and maybe a plate. Picasso probably used the shape of the platter to paint the breast later. The dark shadow around the breast and the crook of her arm loosely follow this underlying dark form. In examining the painting, the conservator could see that there might have been something red, such as a fruit or a flower, under this breast. The X-ray image shows a faint light colored form on this platter, near the highlight of the breast. It may be possible that Picasso adapted the form of the breast from a plate with something on it, a creative choice typical for Picasso. Additionally, some have hypothesized that Picasso saw the curves of a woman in the still-life forms in the first painting; this could have potentially prompted the second painting we see today.

Finally, and perhaps more exciting, it is much easier to see changes in the face, as a result of my methods. Two sets of eyes are visible in the X-ray image and only the lower set appear in the final painting. Also, as the conditioned X-ray image

reveals, the whole head looks like it was slightly more life-like or softer than the mask-like final result. The forehead and hairline seem to be a bit different but it is hard to see due to the flowers. Picasso made these facial changes in the final stages of the painting, well after he abandoned the earlier still-life.

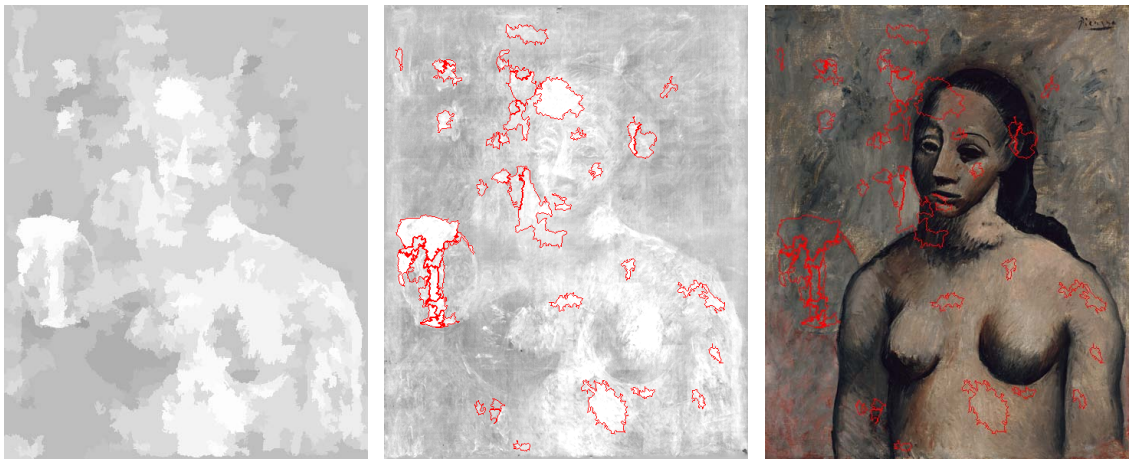
The altered X-ray image more clearly shows the two underlying layers: the still-life, and a different version of the portrait. It clarifies the first composition and then gives us new clues about the development of the final painting. We see Picasso's steps to lay in the still-life and see hints for the underlying inspiration for the portrait, followed by the final facial modifications. Additionally, the X-ray image draws attention to a flower later replaced by the figure's right ear. The conservators had not been able to distinguish these details previously.

Despite these new discoveries, the exploratory visualization tools suffer from a few drawbacks. The most helpful addition to the tools of the conservators is the ability to remove artifacts from the X-ray image. Such artifact removal allows the conservator to see changes previously hidden by the mounting structure of the painting. Although the assumption about the intensity of the X-ray regions relative to luminance-based regions in the first surface of the painting is able to detect several significant features, future methods could employ some more complex transfer functions to help with the identification of shapes that are not just in the thick regions of paint, such as the platter on the left side of the painting, which corresponds to a dark region in the X-ray.

Additionally, some paintings are highly non-uniform; instead of brush strokes, painters in the Pointillist style such as Seurat made tiny dots of vivid color that

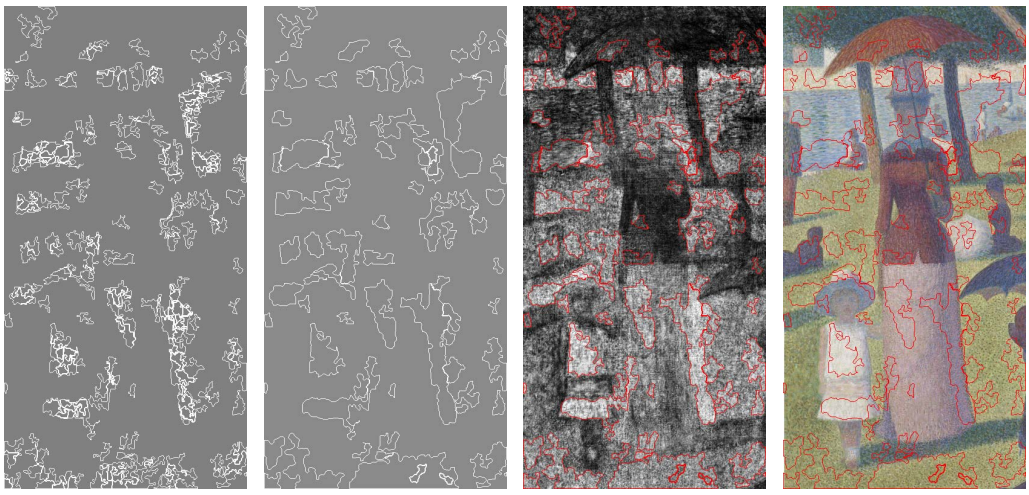
visually merged to give the impression of a uniform area. The artist's method of juxtaposing complementary colored dabs of paint causes problems for mean shift segmentation and creates an X-ray image with a large amount of noise. We found that we can alleviate the problem by applying anisotropic diffusion to carefully blur both inputs.

The mean shift method may cause over-segmentation and produce too many neighboring regions for some images. We simplify these regions by joining the boundaries based upon the color regions they overlap in the segmented color image (Figure 3.4).



(a) Segmented X-ray (b) Mismatched regions on X-ray (c) X-ray Pentimenti

Figure 3.10. Mismatched boundary regions are created from the marked (yellow) pixels by examining each region in the segmented X-ray image where the percentage of marked pixels is above a user specified threshold. These new regions indicate areas in the X-ray image that are statistically different from the first surface of the painting. The images in b) and c) are a result of overlaying the region boundaries both the X-ray image and visible light image.



(a) $X_{boundary}$ (b) Simplified Boundaries (c) X-ray Overlay (d) Visible Image Overlay

Figure 3.11. a) Boundary regions are generated based segmentation of the X-ray image, and may result in many adjoining regions. b) By joining regions in $X_{boundary}$ that overlap color regions in $P_{segmented}$, we can simplify the regions and produce clearer overlays as shown in c) and d).

3.5. Visualizing Pentimenti: Summary

The exploratory data analysis tool for visualizing the pentimenti, or hidden history in paintings, has allowed conservators to remove undesirable artifacts from X-ray imagery. Additionally, we have provided conservators with an initial process for automatically suggesting visually significant regions in the X-ray, aiding in the discovery process. The significant features are much easier to see in the cleaned up X-ray imagery, have enabled many new discoveries, and provided the conservators with a new tool. However, not all of the significant regions are detected from mismatched statistics in images. Automatically finding all of these regions remains a hard problem to solve.

The target audience of the research presented here is conservation scientists, museum visitors, and visualization experts. The tool helps conservation scientists and art historians discover changes, correspondences, similarities and differences in paintings. Specifically, this method of visualizing historic images may help identify areas where the artist changed the painting over time. In order to evaluate the success of the exploratory visualization methods I worked closely with conservation scientists and art historians from the Art Institute of Chicago to effectively display the difference information. I am currently working with other data sets and iterating on the discovery process. In the future, these visualizations could allow museum visitors to interactively explore how artists develop and modify their masterpieces. With the availability of more and more data from UV, IR, and X-ray imaging of artwork, art conservators will benefit from the efforts of the visualization community to help explore and analyze these very large and rich data sets.

CHAPTER 4

Conclusion

This dissertation claims and provides supporting arguments and evaluation for two contributions to the computer graphics community. First, image creation and manipulation should be primarily concerned with examining perceptual differences over regions larger than individual pixels. Second, once we have calculated these differences, we can map changes from one domain to another, while preserving and often enhancing the communication content of the imagery. In Chapters 2 and 3, provides support and evaluation of these two contributions via two problems: a color-to-grayscale conversion algorithm and a method for comparing and analyzing a visible light image and an X-ray image of historically significant paintings.

4.1. Concluding Remarks on Color2Gray

Chapter 2 presents a Color2Gray method that examines large pixel neighborhoods and selectively incorporates the original luminance and color differences in a new grayscale image. A dimensionality reduction technique such as any color-to-gray algorithm, reduces an image from tristimulus values to a single channel, and ultimately results in a loss of information. However, by preserving relative differences in both chrominance and luminance, the Color2Gray algorithm maintains salient features from a color image in the final grayscale result. The results

of this method show that careful image manipulations that preserve perceptual differences create improved grayscale images and are useful when it is not possible to reproduce a full color image, such as on a grayscale printer. In addition to being useful for improving grayscale printer drivers, the techniques used in our Color2Gray method may be useful for detecting camouflage or possibly even creating better camouflage.

Although default parameters for the Color2Gray method allow automation, the level of salience and semantic preservation often depends upon the choice of parameters and the three simple parameters of the algorithm can easily be tuned by a viewer based upon the content of the image. For example, the sunrise image in Figure 2.1 makes the most sense when the sun is brighter than the background, but a default setting of the parameters could cause a reversal of the intensities and confound the semantic mapping.

From the images examined in this work, it appears that the Color2Gray method is most useful for images with high color contrast but low intensity contrasts, such as images that capture computer-generated renderings or artworks on canvas. Further investigations should include the analysis of the effect and usefulness of the algorithm on natural images; perhaps different techniques would be necessary to emphasize the subtle variations found in natural images, although this problem becomes quite similar to high-dynamic range imaging approaches.

Limitations and areas of future research for the Color2Gray algorithm include how to selectively choose luminance changes over chrominance changes, how to

preserve values of pixels or regions in an image, as well as how to create parameters which allow region-based control of color mapping, contrast, and which pixels effect resulting grayscale values.

Further investigations of the algorithm should include an evaluation of the choice of the forced-choice selection of ΔL_{ij} over $||\overrightarrow{\Delta C}_{ij}||$ as described in Section 2.3.2. The discontinuous case worked best because linear or other blending functions resulted in muddier images because when the two terms have opposite signs, the differences cancel out. However, many other functions could be evaluated.

Some source images may require that certain intensity relations be preserved, but the techniques presented here can not guarantee that the intensity of a given pixel or region will stay fixed and some regions may be pushed in the opposite direction to account for chrominance shifts. Although, the algorithm starts with the default grayscale image, and the new grayscale conversion will not differ by much, users might want controls to pin down specific values in an image and allow the rest to adapt.

Additionally, the algorithm currently does not allow for multiple settings for parameters over different regions in the image, and thus does not allow control over the relational mapping of color pairs (θ) or other side effects such as contrast (α), or the size, shape and magnitude of region affected by neighboring pixels (μ).

Future algorithms may also include the ability to map color to semantic features such as glyphs or texture to aid in the identification of regions in grayscale. For example, a grayscale reproduction of a simple x-y plot with multiple lines may be

better communicated with different textures or symbols instead of different shades of gray if color is not available.

4.2. Concluding Remarks on Visualizing Pentimenti

Chapter 3 presents a series of techniques for identifying the hidden history, or pentimenti, of a work of art by examining an aligned pair of images in the form of a visible light image and an X-ray image. First, I present a method for helping conservators selectively edit X-ray images to remove artifacts such as mounting structures and X-ray mosaic seams by editing changes in the gradient domain and rebuilding the image with a Poisson solver. Next, because sensible pixel-to-pixel comparisons between images in these two different domains (X-ray and visible light) are not possible, I look at statistical correlation across regions. By examining coarse regions instead of pixel values, it is possible to establish statistical correspondences between changes in the density of paint indicated by changes in the X-ray image and segmented regions from the luminance channel of the visible light image. The resulting images and interactions enable conservators and other domain experts to further examine clues with regard to the development of a masterpiece of art. As a result of this work, conservators at the Art Institute have made several new discoveries and these techniques will be used to examine more digital archives of paintings in the near future.

Although it may have been possible to remove most of the X-ray artifacts with discrete frequency-domain analysis using the FFT (Fast Fourier Transform) or related methods such as editing wavelet coefficients, automatic techniques are not

desirable due to their indiscriminate removal of data. Conservators at the Art Institute of Chicago prefer to have control over what seams and artifacts are removed, and hence require interactive editing for artifact removal. Additionally, the current methods for marking important regions proved to be marginally helpful. The domain experts, although excited about the artifact removal, would much rather see the marking and removal of corresponding regions in order to emphasize the parts of the X-ray which do not match the first surface of the painting. Although the techniques provided in this dissertation will easily allow the marking of corresponding regions, the removal of such regions is nontrivial. It may be possible however to use gradient-domain techniques similar to the artifact removal to reduce the appearance and eventually fade-away regions which correspond to the first surface of the painting.

4.3. Closing Comment

While pixel comparisons are common, they are insufficient for many image creation and manipulation tasks. In my research, I claim that by examining differences between regions in an image, new mappings, not restricted to one-to-one mappings, enable new opportunities for effectively conveying information. Novel mappings other than one-to-one photorealism must be done carefully to create informative and aesthetically pleasing images. Additionally, interaction is necessary to preserve and convey the salience of images altered through image processing. Both of the methods presented in this dissertation maintain a balance between the level of human interaction and the level of automation.

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APPENDIX A

Additional Color2Gray Images

It is important to note that all of the images in this appendix were generated with a previous version of the Color2Gray code, in which we had a different algorithmic description for $theta$. In this document: $\theta_{appendix}$ = θ + 90°. All other operations in the code are as represented in Chapter 2 of this dissertation.

A.1. Contrast-based Attention Model Analysis

As shown in Figure A.2, we conducted a preliminary evaluation of our Color2Grey algorithm via a contrast-based image attention model by Ma and Zhang [**31**]. These results indicate that the Color2Gray results maintain the visual salience characteristics of the original color images on the average better than Adobe Photoshop's grayscale methods.

A.2. Varying θ parameter

Each of the wheel-like illustrations in Figures A.3 - A.12, illustrate the Color2Gray algorithm applied to the color image in the center of the figure over several values of the parameter θ. The dot product of this vector with the vector between the chrominance values of any two pixels, i and j, determine the sign of the luminance change $C[i, j]$.

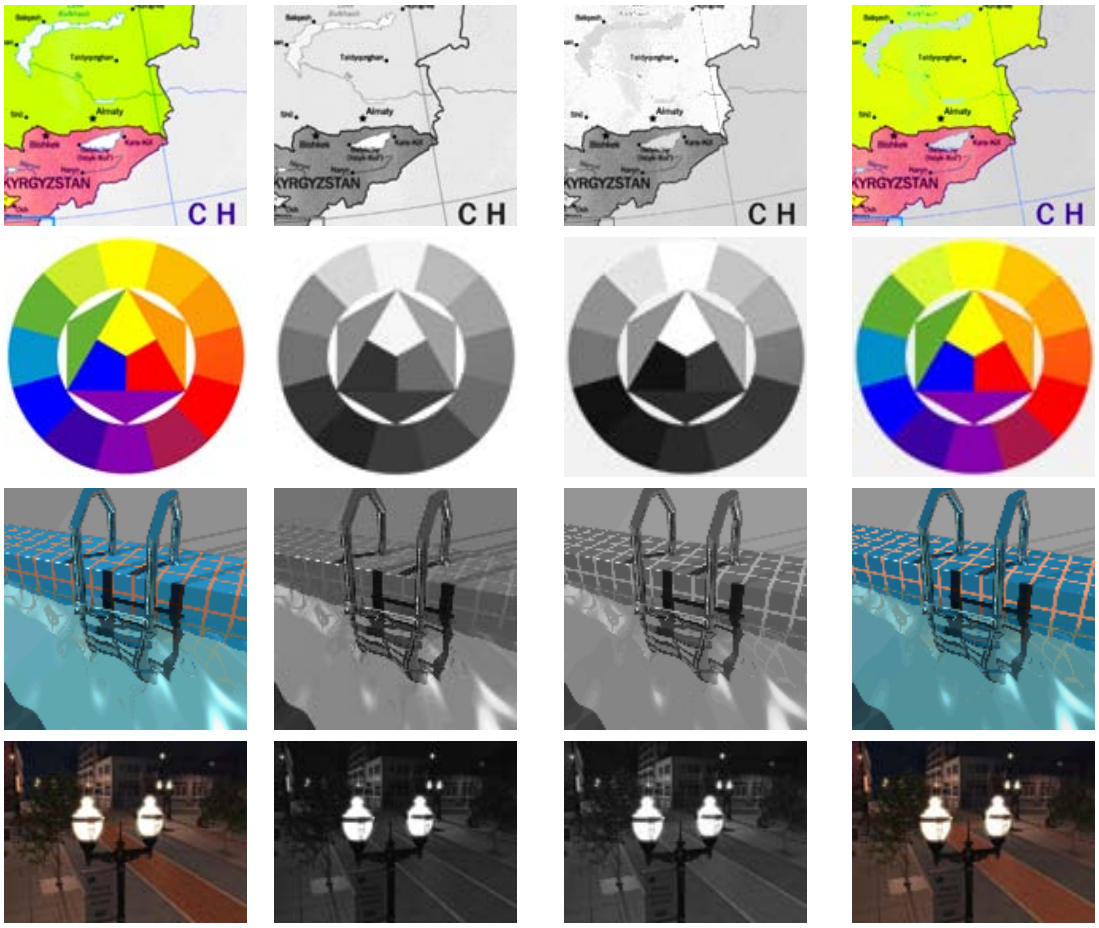

Figure A.1. Several results, comparing of Original Color image, Traditional Grayscale, and Color2Gray Results, and a Color2Gray + chrominance channels image.

A.3. Images from the paper with new color images

Figures A.13 and A.14 provide more examples of creating "color+" images. We can create a separate “color+" image at the same time we create the new grayscale images by simply adding back the chrominance channels before converting the image back to RGB. This may be useful for creating full color images with a perceptually salient luminance channel. For example, one could create a graph for a paper

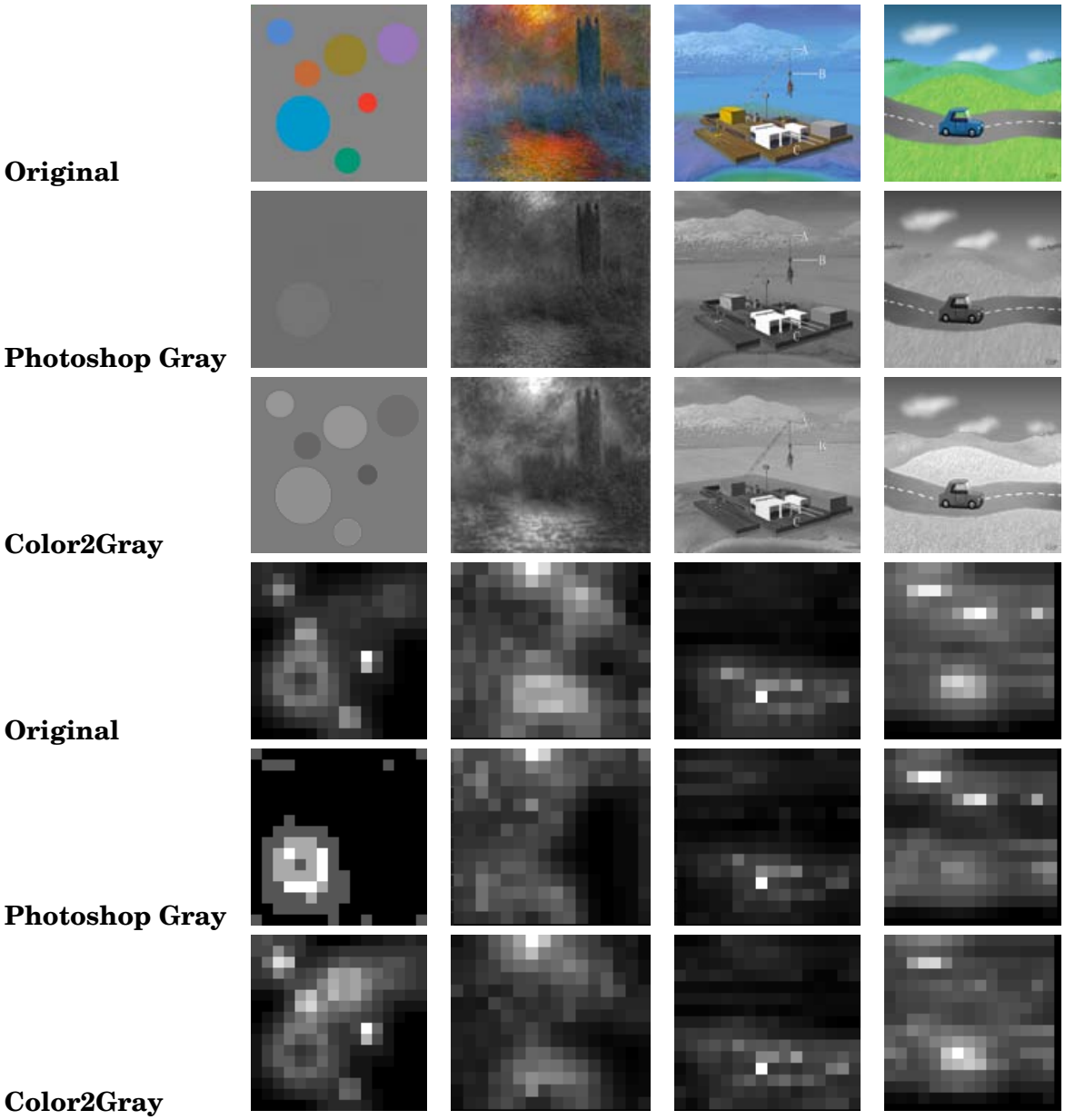

Figure A.2. Contrast-based image attention model [Ma and Zhang 2003] applied to original image, Photoshop grayscale, and Color2Gray images.

which will print well in color and grayscale. Additionally it provides useful feedback for visualizing the subtle adjustments made with the Color2Gray algorithm.

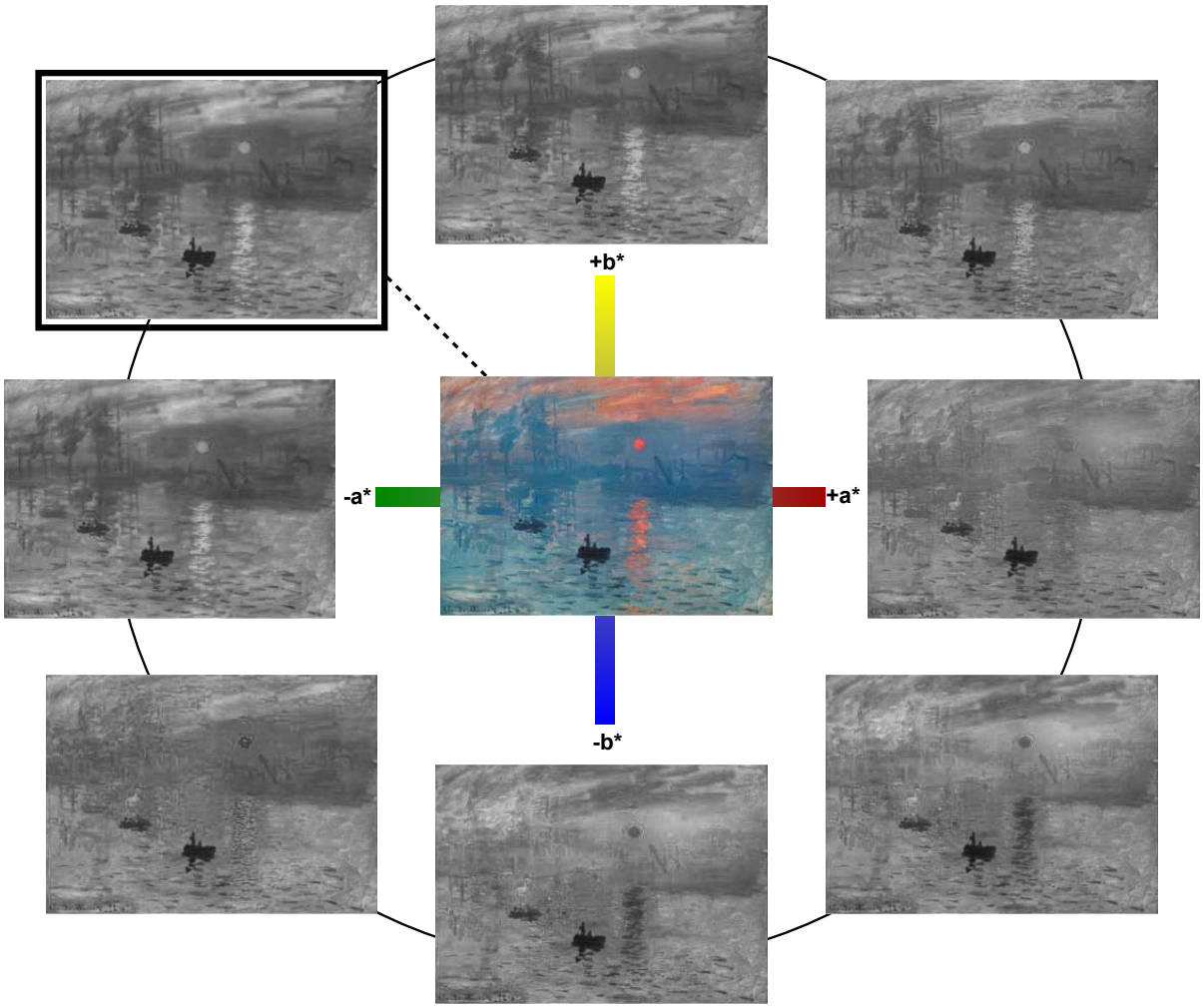

Figure A.3. Color2Gray results for the color image in the center of the figure over several values of the parameter θ.

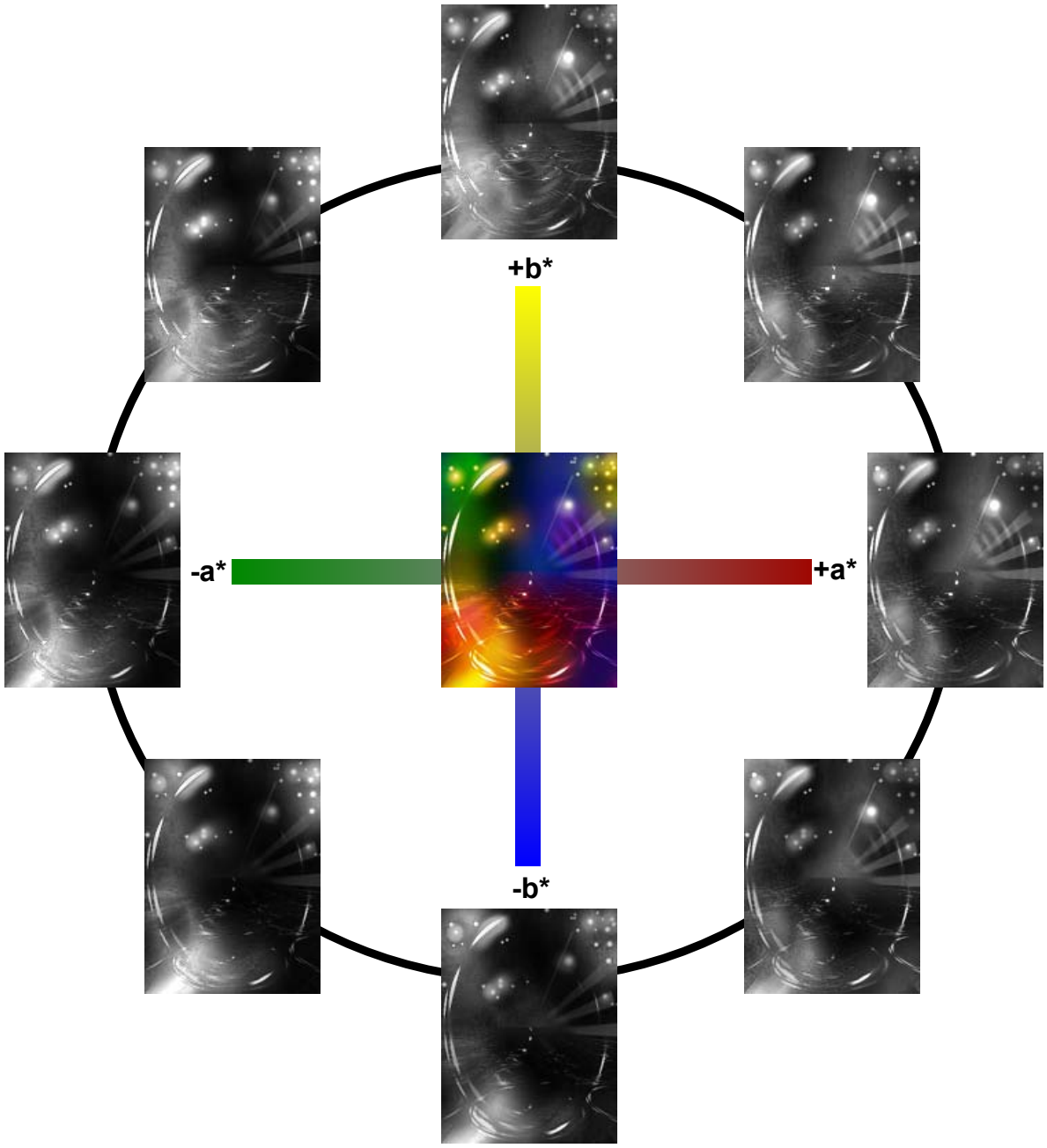

Figure A.4. Varying the parameter θ results in subtle changes and sometimes it may not be clear which setting produces the best grayscale representation of the original color image. Image courtesy of Ana Vasileva.

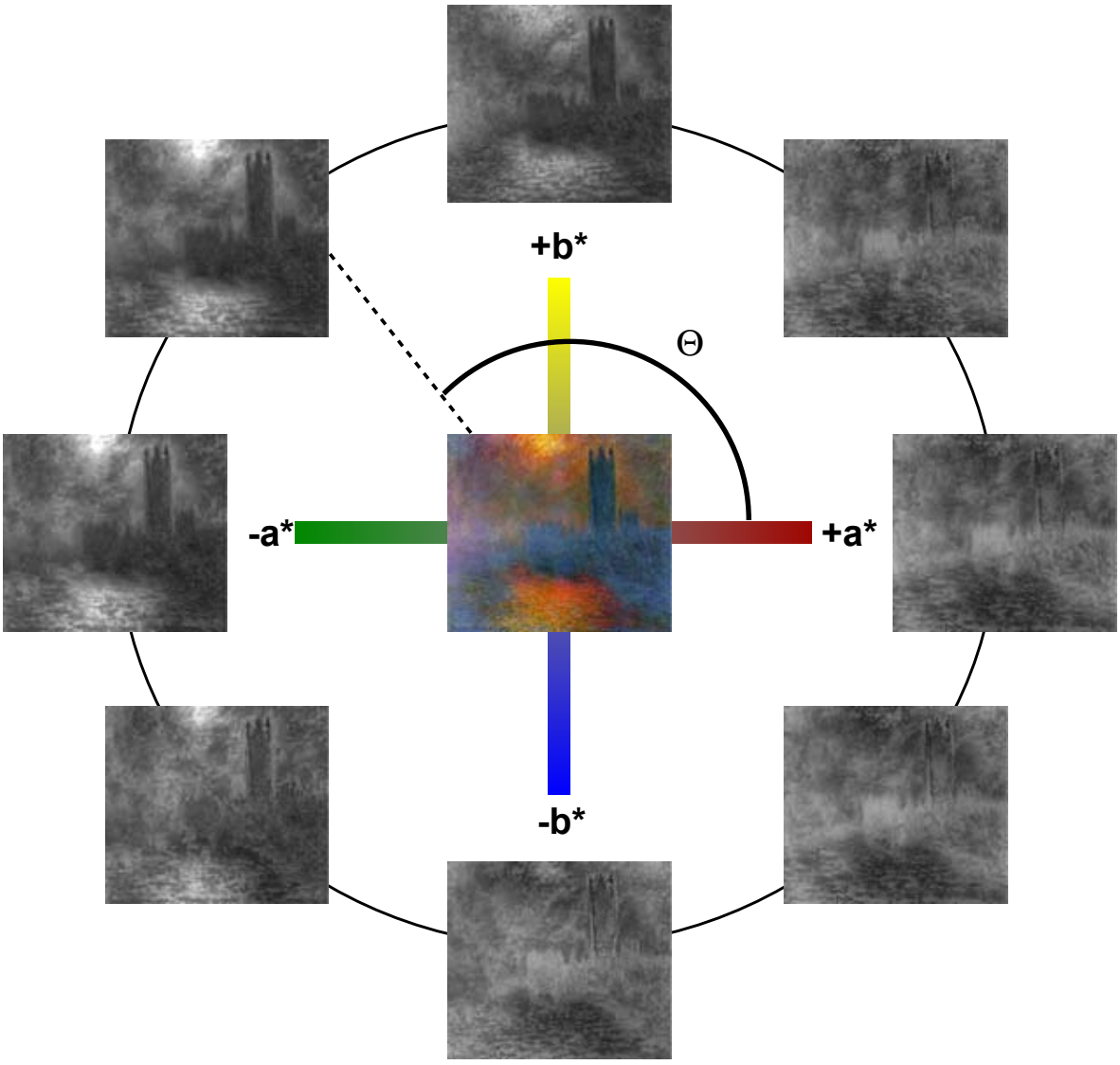

Figure A.5. Varying the parameter θ can sometimes lead to "inverted" images. Painting by Claude Monet.

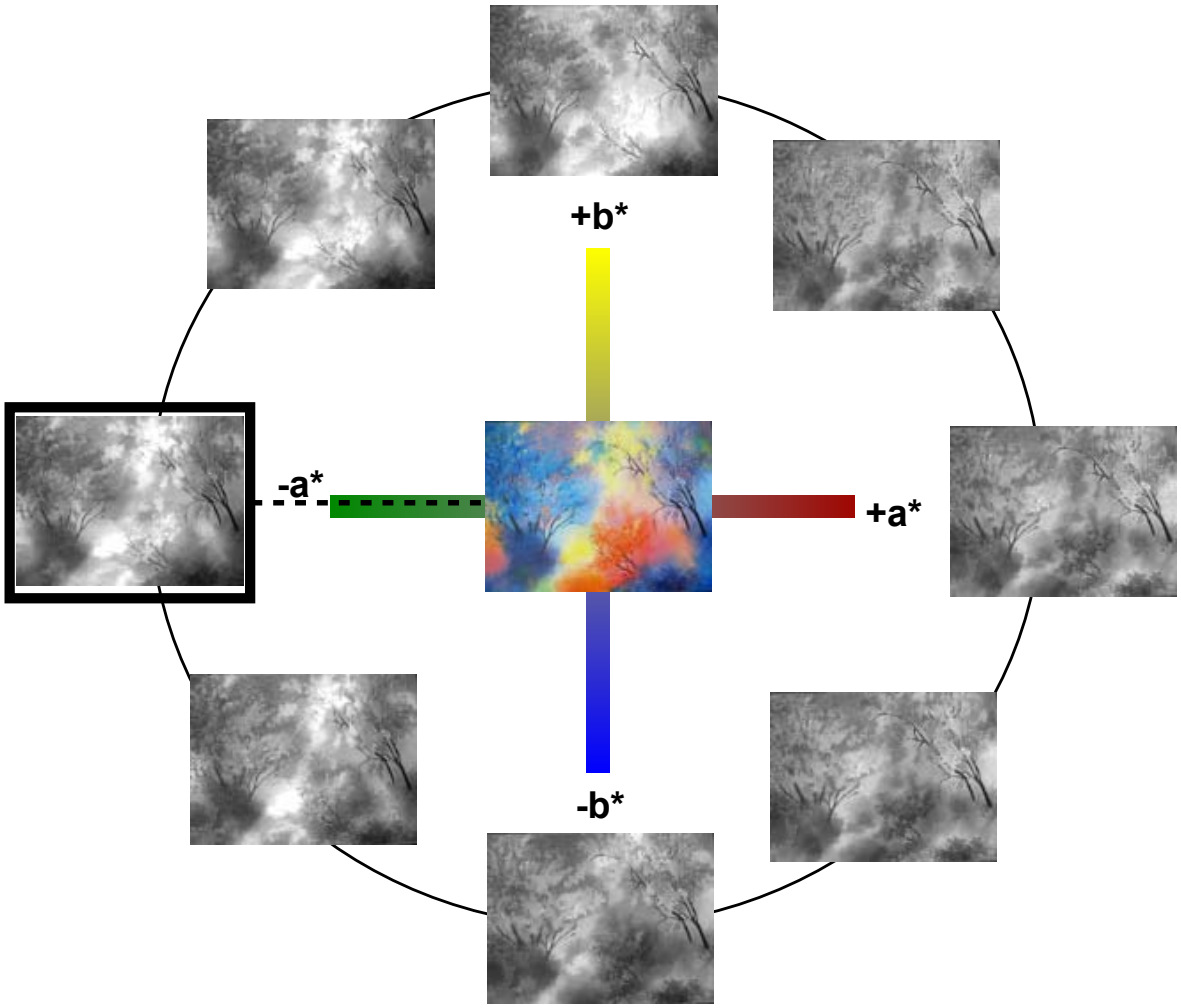

Figure A.6. It is difficult to pick which of these images would be the ideal grayscale image.

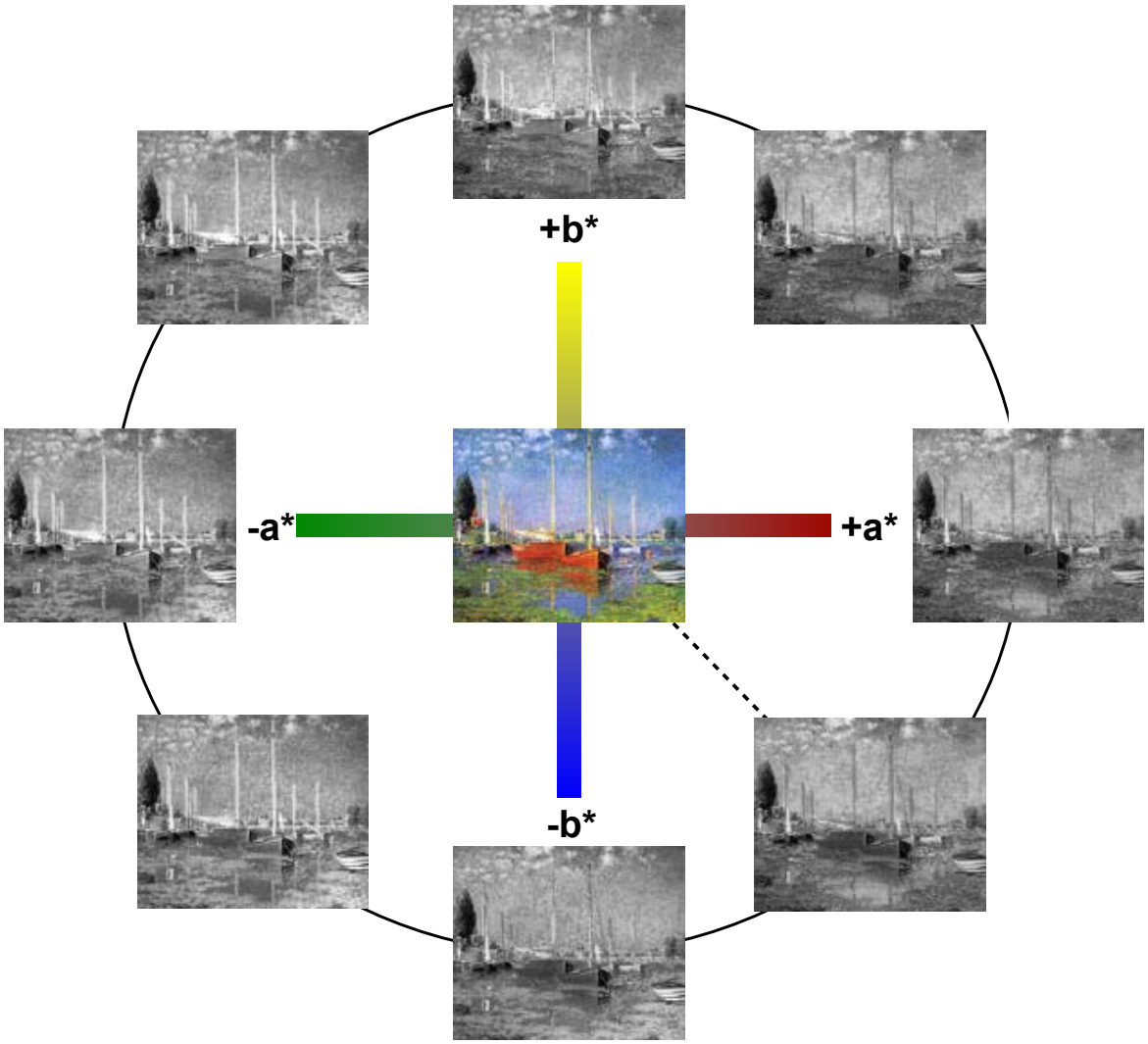

Figure A.7. In this sail boat image by Monet, the Color2Gray results help to differentiate the plants on top of the water from the water.

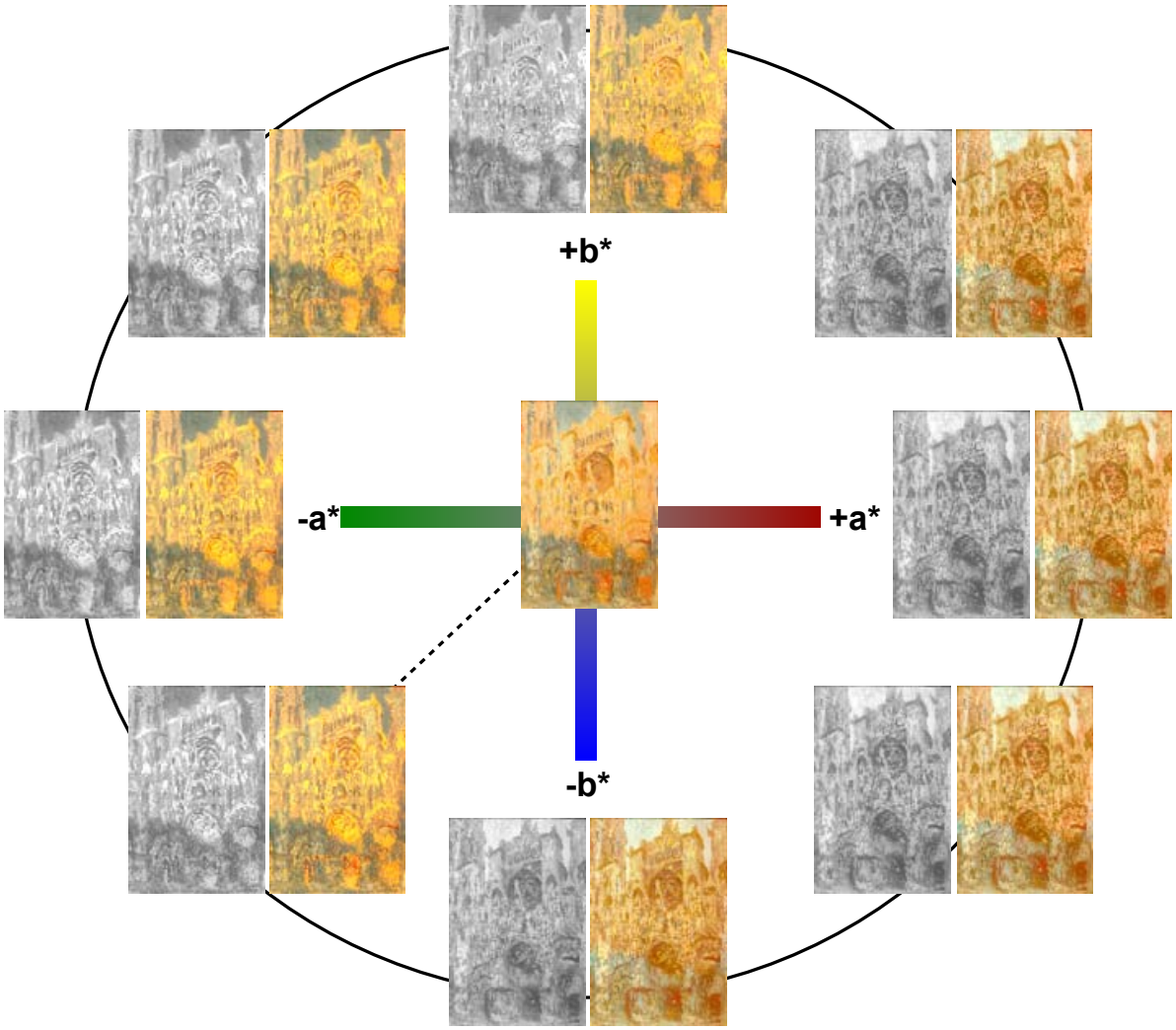

Figure A.8. Color2Gray and recolored-Color2Gray image pairs over a range of θ values. Painting by Monet.

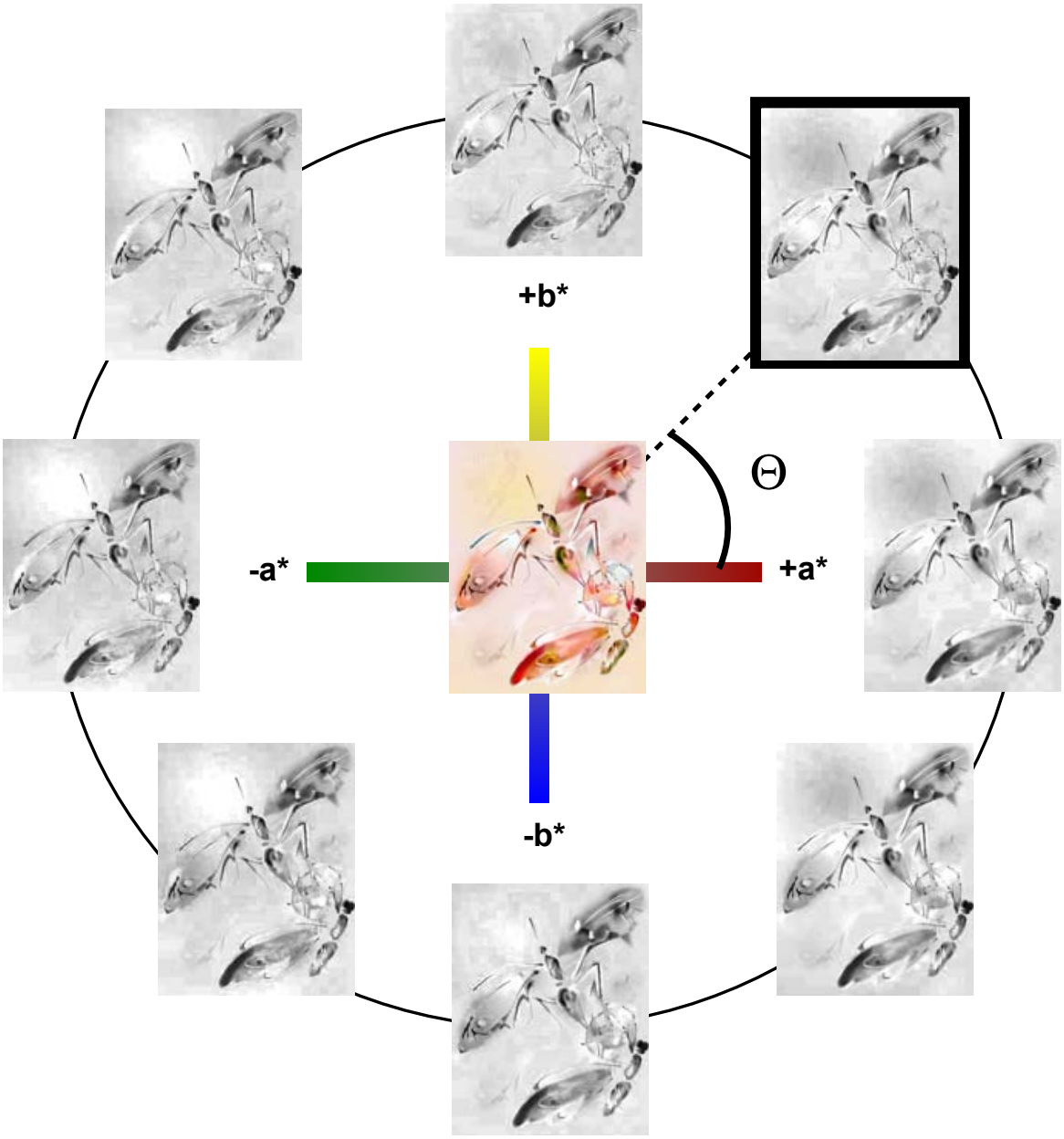

Figure A.9. Varying θ can subtly alter the semantic meaning of the yellow fade found in the original color image behind the head of the butterfly. Image courtesy of Ana Vasileva.

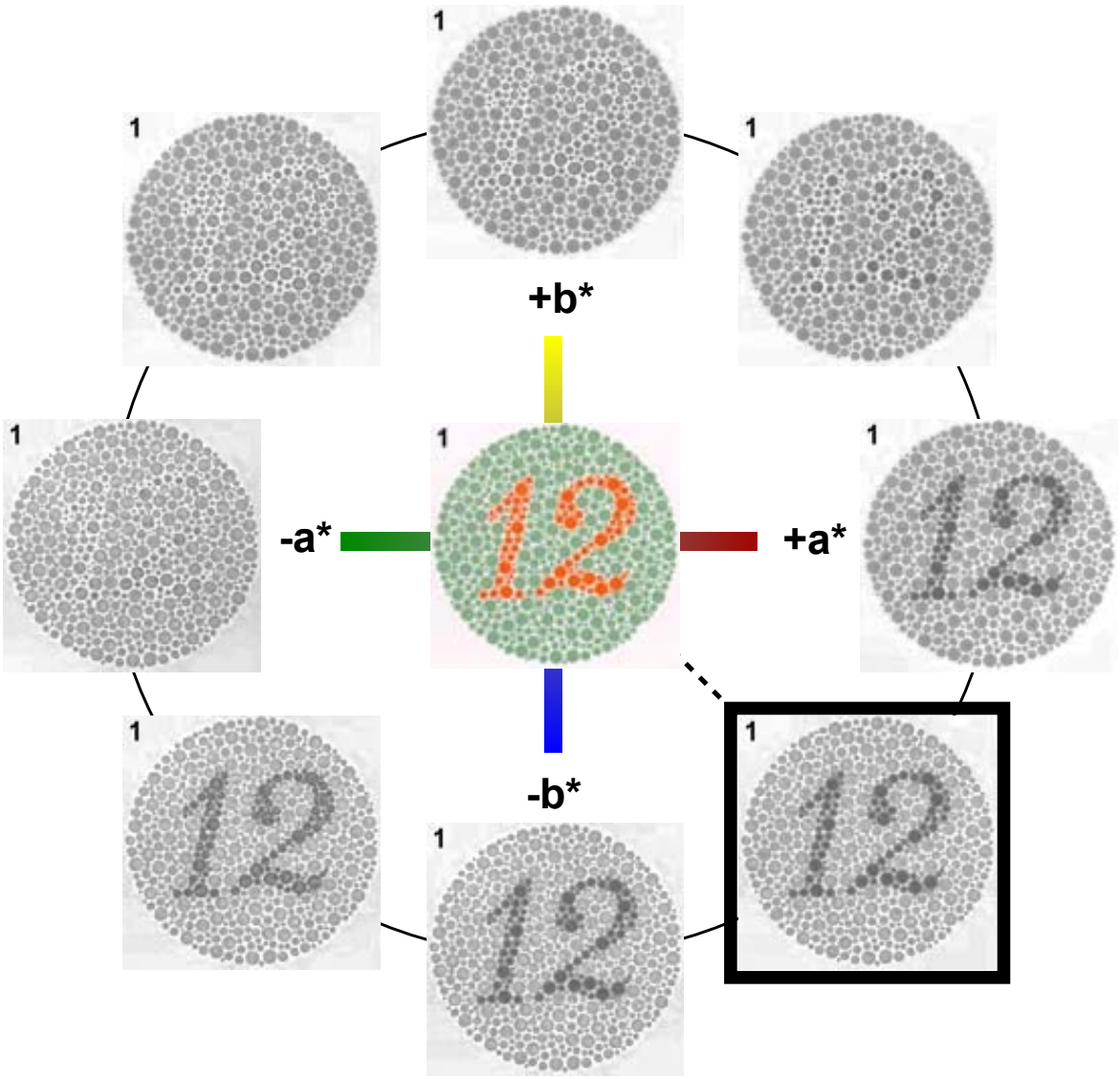

Figure A.10. Varying parameter θ on color blindness test image with number 12. Image copyright Jay Neitz.

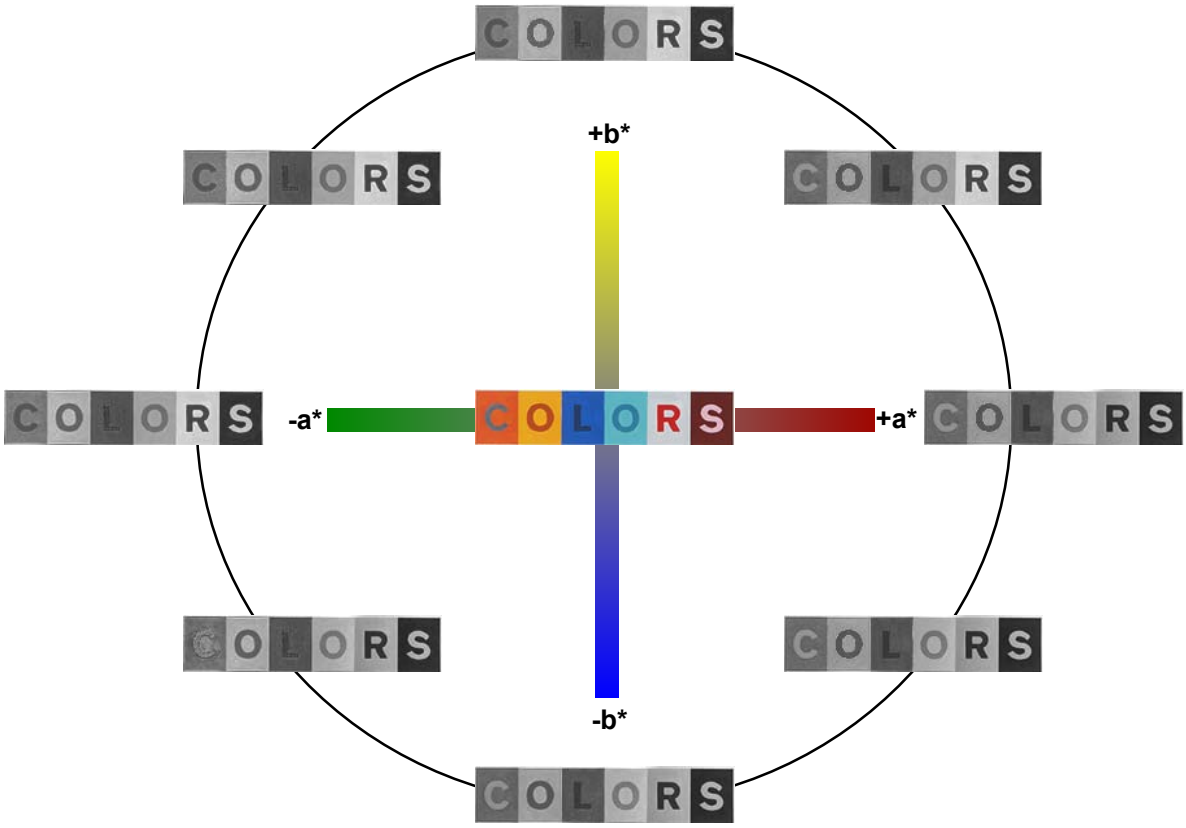

Figure A.11. Strip of colors converted to grayscale with a range of θ values.

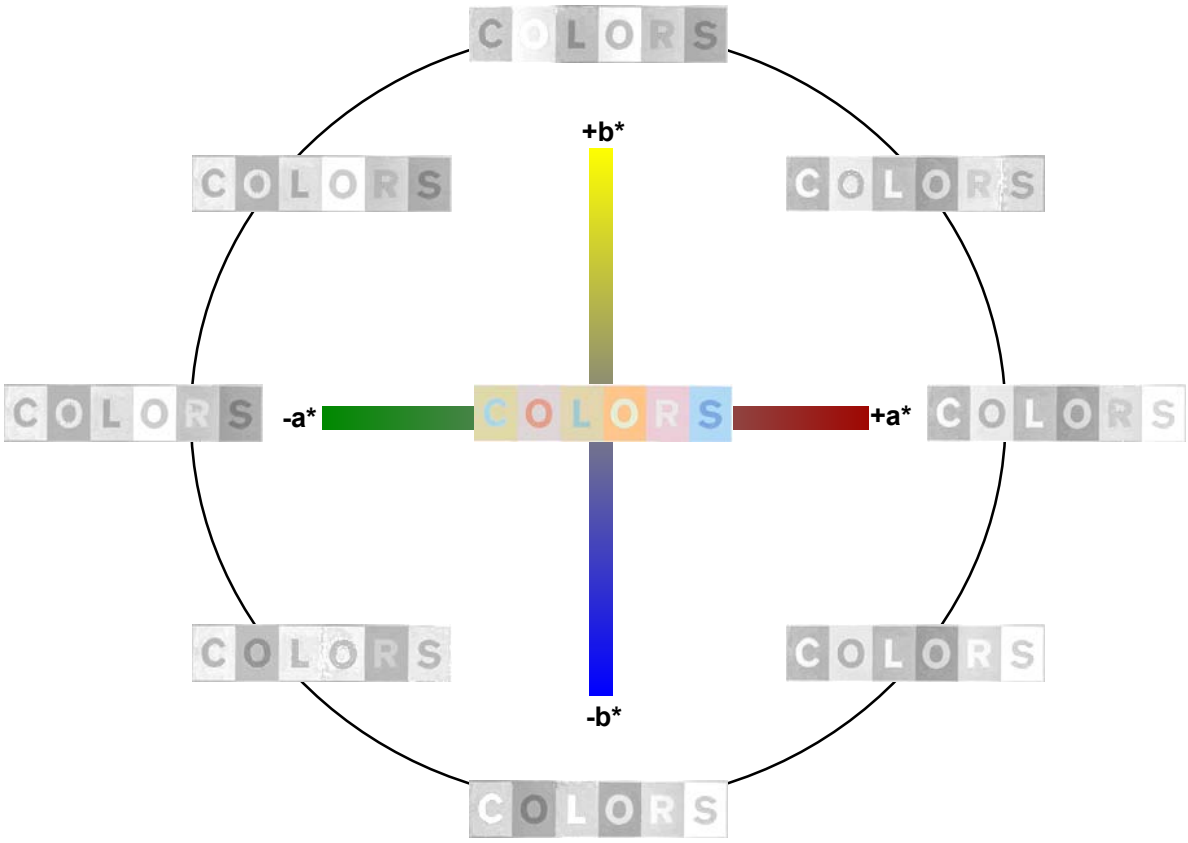

Figure A.12. Another strip of colors converted to grayscale with a range of θ values.

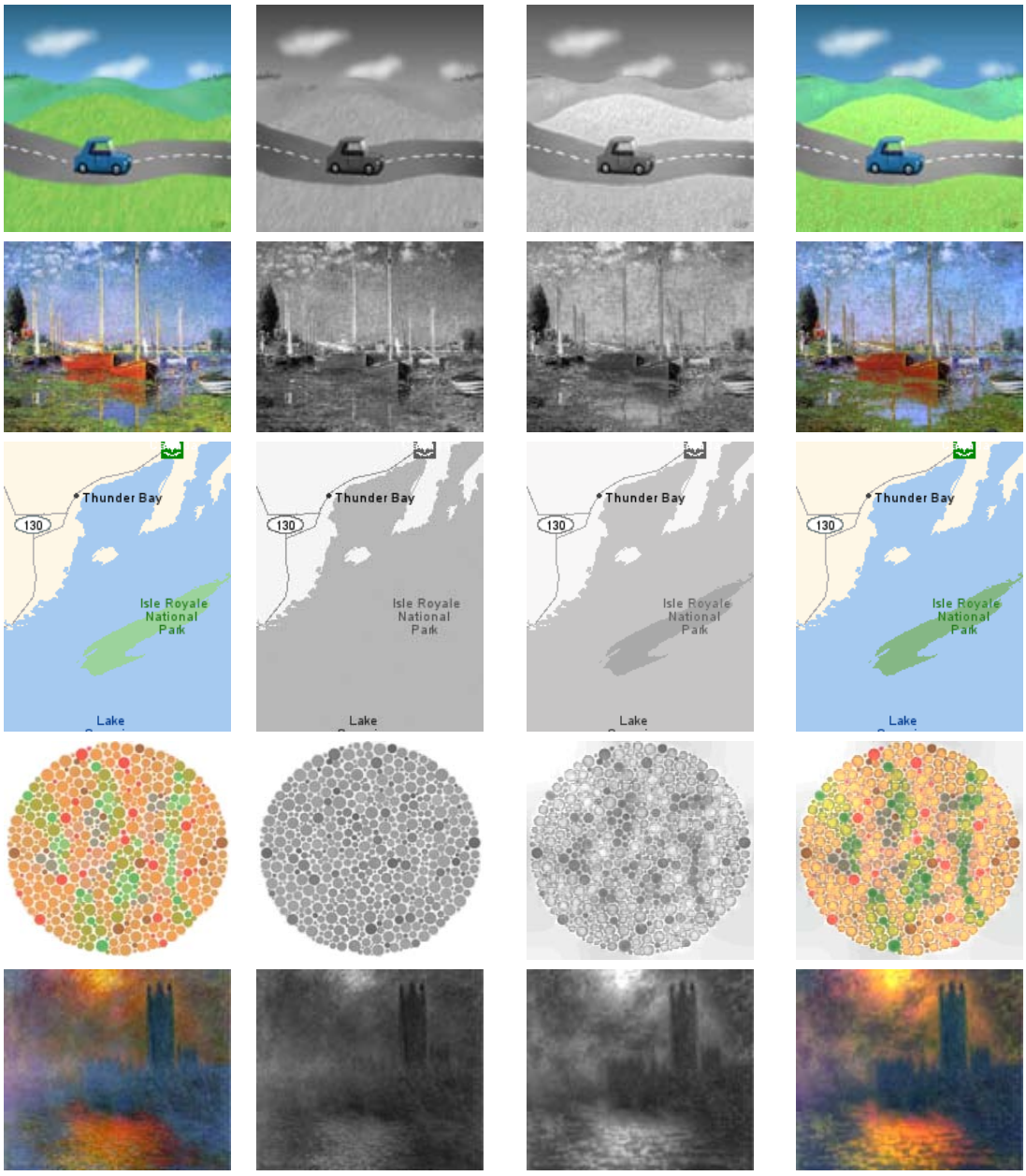

Figure A.13. Comparison of Original Color image, Traditional Grayscale, and Color2Gray Results, and a Color2Gray + chrominance channels image.

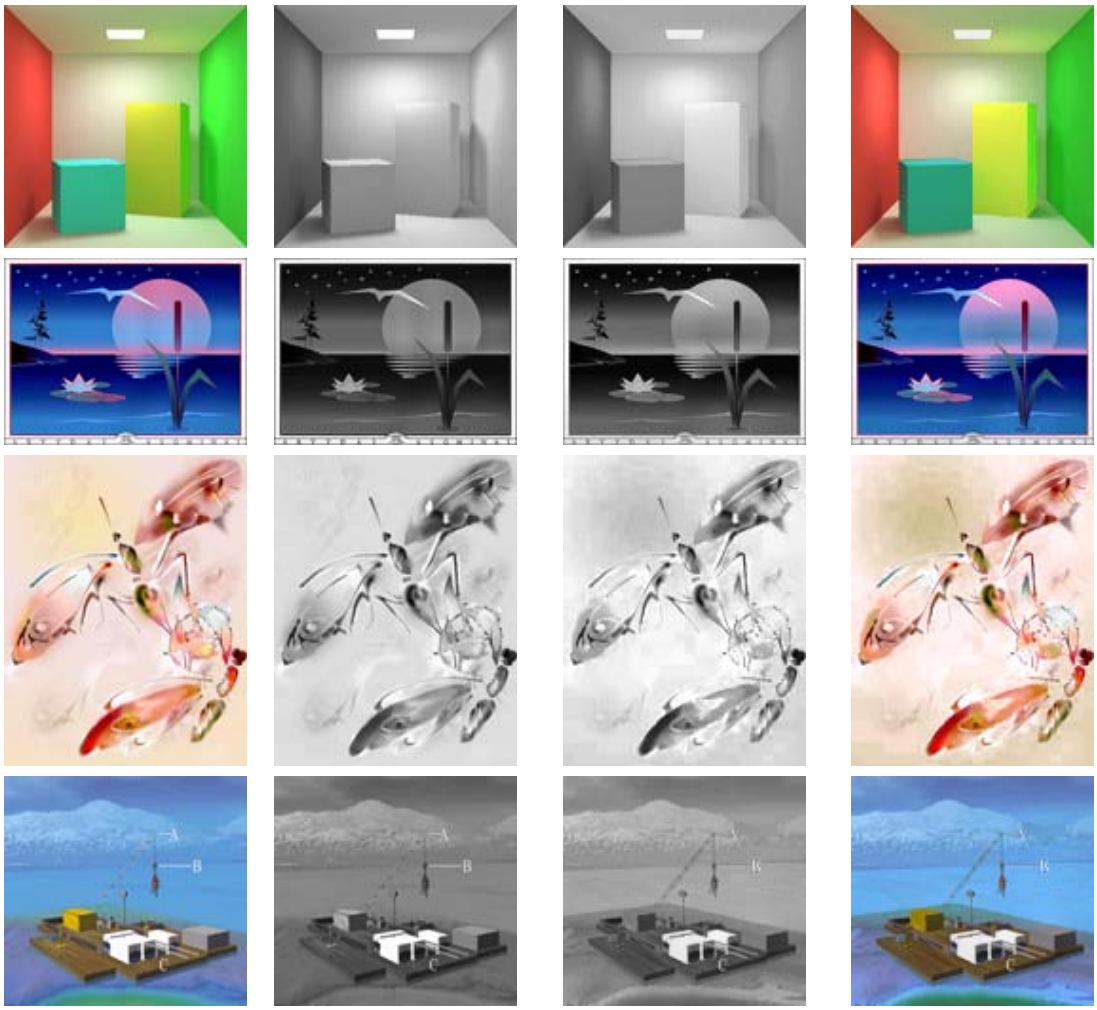

Figure A.14. Comparison of Original Color image, Traditional Grayscale, and Color2Gray Results, and a recolored-Color2Gray image.

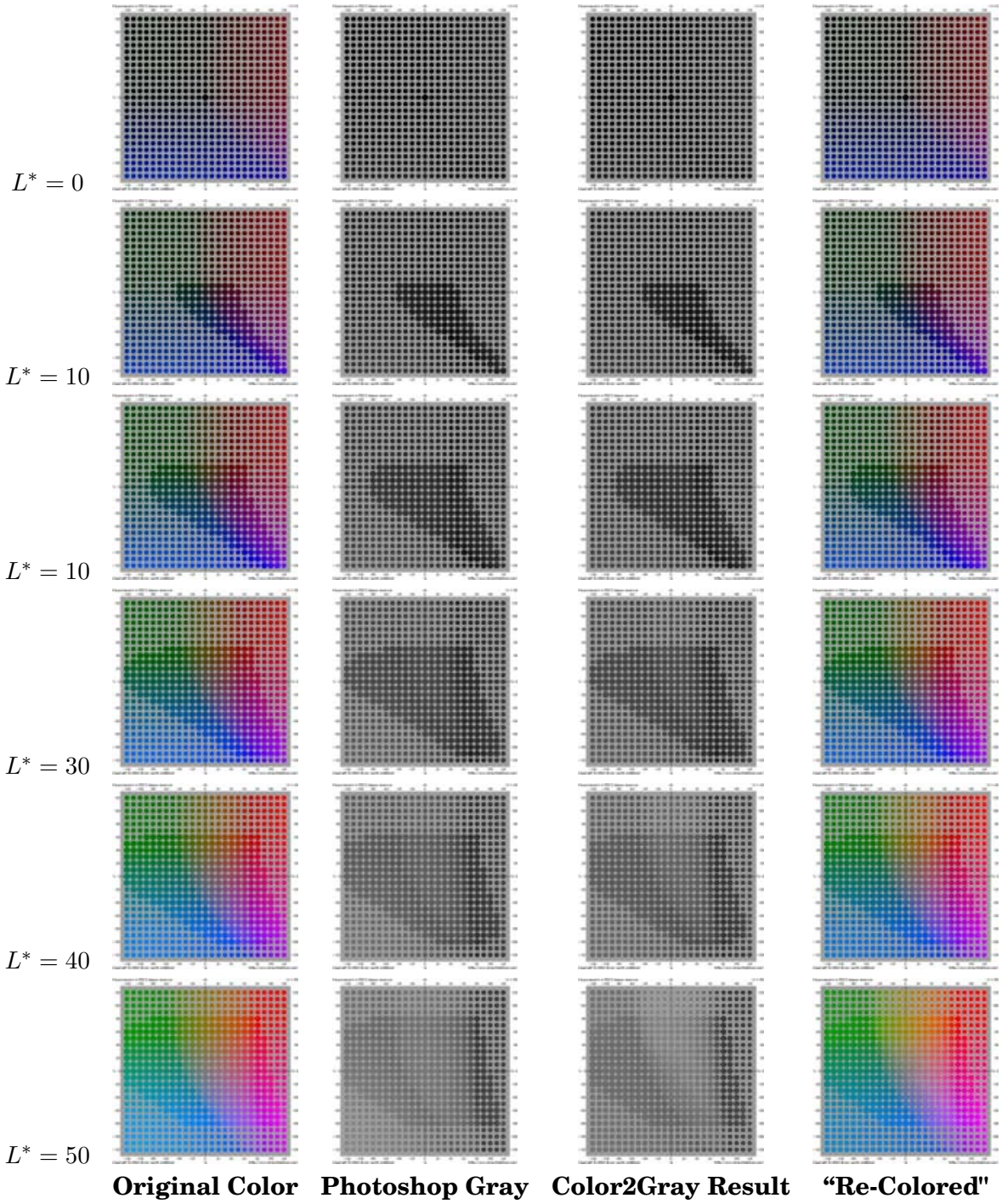

Figure A.15. Color2Gray ($\theta = 45°$, $\alpha = 8$) applied to LAB Profile Plots from Bruce Lindbloom, `http://www.brucelindbloom.com/index.html?UPLab.html`.

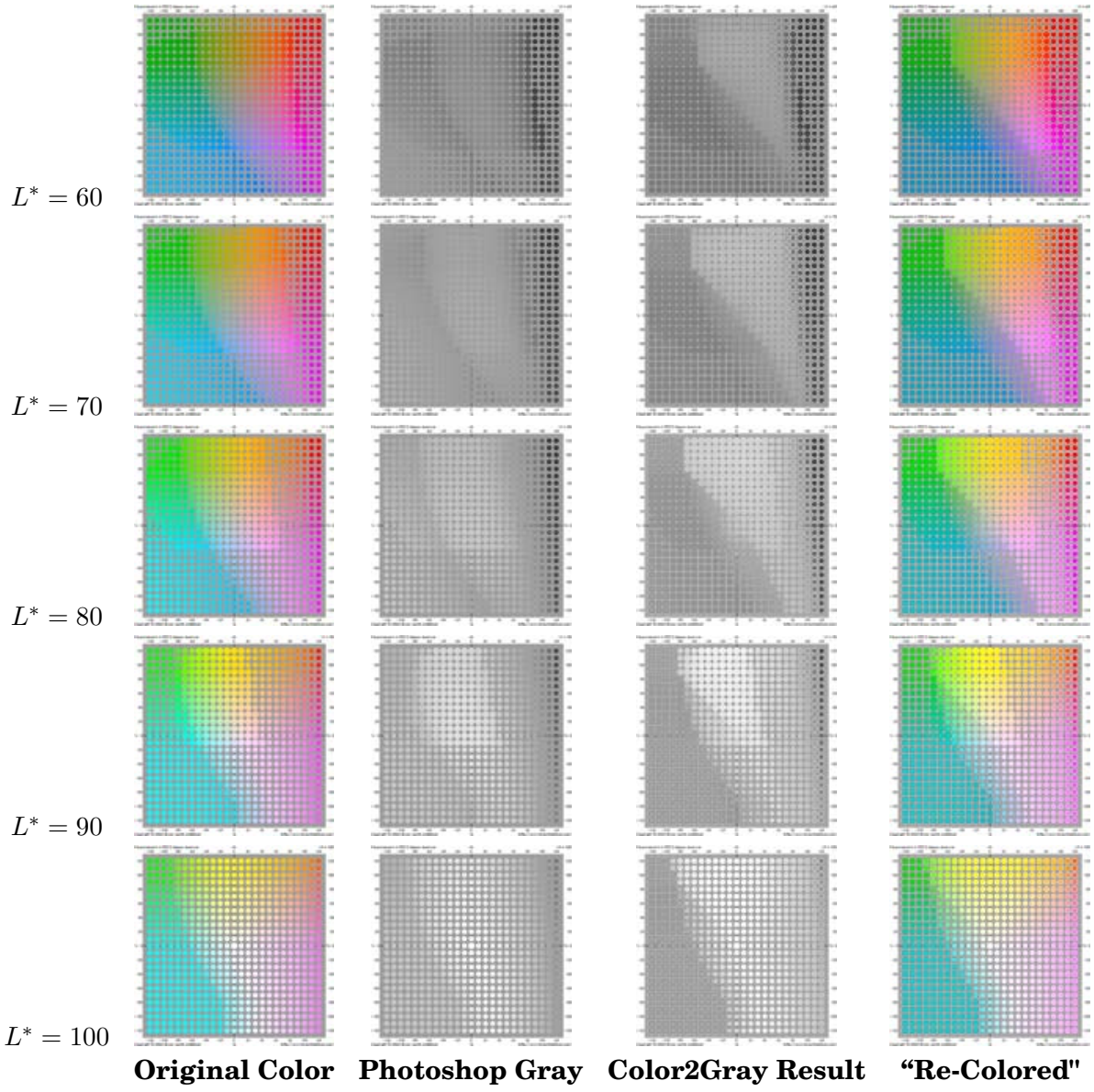

Figure A.16. Color2Gray ($\theta = 45°$, $\alpha = 8$) applied to LAB Profile Plots from Bruce Lindbloom, `http://www.brucelindbloom.com/index.html?UPLab.html`.